

Mathematics for Data Science 1

Comprehensive Textbook & NPTEL Lecture Companion

IIT Madras BS in Data Science and Applications

Prepared for Beginner-Friendly Learning, Deep Concept Mastery, & Data Science Application

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Introduction

Welcome to the **Mathematics for Data Science 1 (MDS1)** Comprehensive Textbook. This document is designed to serve as a complete, beginner-friendly resource to help you learn, practice, and master every single topic covered in the IIT Madras Foundational Level course.

Structure of this Guide

The book is divided into **12 Chapters**, corresponding to the 12 weeks of the official course curriculum:

- **Chapters 1–4 (Quiz 1 Content):** Sets, Relations, Functions, Coordinate Geometry, Straight Lines, Quadratic Equations, and Polynomial Algebra.
- **Chapters 5–8 (Quiz 2 Content):** Functions, Logarithms, Exponential Equations, Sequences, Limits, Continuity, and the foundations of Differentiation.
- **Chapters 9–12 (End-Term Content):** Integration (area under a curve), and an extensive, algorithm-focused dive into Graph Theory and Shortest Path/MST Algorithms.

Key Design Elements

Throughout this textbook, key ideas are categorized in visually distinct, colored cards:

- **Teal Cards** contain definitions, terminology, and core concepts.
- **Slate Blue Cards** contain theorems, rules, and mathematical properties.
- **Purple Cards** contain detailed step-by-step solved examples.
- **Orange Cards** highlight crucial formulas you must commit to memory.
- **Video Resource Cards** provide direct clickable links to official IIT Madras NPTEL lecture videos and handpicked visual explanation videos.

Chapter 1

Sets, Relations, and Functions

This chapter covers the foundational building blocks of mathematical language: number systems, sets, relations, and functions. While these topics might seem entirely abstract at first glance, they are the very same tools used to formalize data structures, database tables, and the mapping of inputs to outputs in machine learning models.

1.1 Number Systems

Before we can organize items into sets, we need to establish the standard sets of numbers used in mathematical analysis. In data science, understanding number systems helps you decide how to represent features in memory (e.g., integers vs. floating-point numbers) and dictates what kind of statistical models you can apply.

- **Natural Numbers ($\mathbb{N}$):** $\mathbb{N} = \{1, 2, 3, \dots\}$. These are counting numbers. In a dataset, the number of clicks on a website or the number of words in an email would be represented using natural numbers. (Note: Some contexts include 0).
- **Integers ($\mathbb{Z}$):** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$. Integers include negative values. We often use integers to represent changes in stock prices (up or down by whole amounts) or differences in counts over time.
- **Rational Numbers ($\mathbb{Q}$):** $\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$. These are numbers with terminating or repeating decimal representations.
- **Real Numbers ($\mathbb{R}$):** The set of all rational and irrational numbers (such as $\sqrt{2}, \pi, e$). $\mathbb{R}$ represents the entire continuous line of numbers. In machine learning, continuous features like temperature, weight, or calculated probabilities (between 0 and 1) are treated as real numbers.

Data Science Relevance: Knowing the domain of your data is the first step in exploratory data analysis. A machine learning algorithm handles a column of Integers ($\mathbb{Z}$) representing categorical IDs very differently from a column of Real numbers ($\mathbb{R}$) representing house prices.

Video Resource: IITM BS Lecture: W1_L3 Real & Complex Numbers



Official IIT Madras lecture explaining rational and irrational numbers in depth.

1.2 Set Theory

A **set** is a well-defined, unordered collection of distinct objects. The objects contained within a set are called its *elements* or *members*. Sets are usually denoted by uppercase letters (A, B, C) and elements by lowercase letters (a, b, x, y).

Definition: Sets and Subsets

Let A and B be sets.

- **Element of:** $x \in A$ means x is an element of set A .
- **Subset:** $A \subseteq B$ (A is a subset of B) if every element of A is also in B :

$$\forall x (x \in A \implies x \in B)$$

- **Proper Subset:** $A \subset B$ if $A \subseteq B$ and $A \neq B$. This means B has at least one element that is not in A .
- **Empty Set ($\emptyset$):** A set containing no elements. Interestingly, $\emptyset \subseteq S$ for any set S .
- **Power Set ($\mathcal{P}(S)$):** The set of all subsets of S . If $|S| = n$ (meaning S has n elements), then the power set has $|\mathcal{P}(S)| = 2^n$ elements.

Data Science Relevance: In Python, the 'set' data structure is used to quickly remove duplicates from a list (e.g., finding all unique words in a document). A dataset itself can be viewed as a set of data points, and finding a subset (like filtering rows where 'age > 30') is a fundamental data manipulation operation.

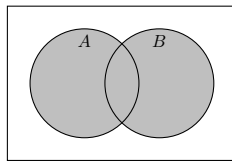
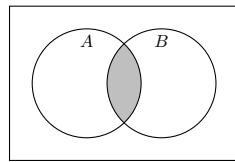
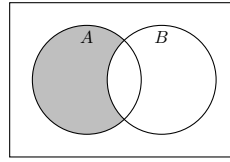
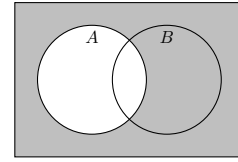
Video Resource: IITM BS Lecture: W1_L4 Set Theory 

Official IIT Madras lecture introducing sets, subsets, and fundamental operations.

1.2.1 Set Operations

Just as we can add and multiply numbers, we can combine sets using specific operations. For any universal set U (the set containing everything under consideration) and subsets $A, B \subseteq U$:

- **Union ($A \cup B$):** Elements in A or B (or both). Logically equivalent to the "OR" operation.
- **Intersection ($A \cap B$):** Elements in both A and B . Logically equivalent to the "AND" operation.
- **Difference ($A \setminus B$ or $A - B$):** Elements in A but not in B . This is like saying "Give me users who bought a laptop (A) but did NOT buy a mouse (B)."
- **Complement (A^c or A'):** Elements in the universal set U that are not in A .
- **Cartesian Product ($A \times B$):** The set of all possible ordered pairs where the first element is from A and the second is from B : $\{(a, b) \mid a \in A, b \in B\}$.

Union $A \cup B$ Intersection $A \cap B$ Difference $A \setminus B$ Complement A^c **Example: Set Operations in Python**

In Data Science, we routinely perform these exact mathematical operations using Python's set object:

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3
4 union = A | B           # {1, 2, 3, 4, 5}
5 intersection = A & B    # {3}
6 difference = A - B      # {1, 2}
7 sym_diff = A ^ B        # {1, 2, 4, 5}
```

Theorem/Rule: Laws of Set Algebra

For any sets A , B , and C :

- **De Morgan's Laws:**

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

- **Distributive Laws:**

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

```
1 # Define sets of user IDs who bought laptops vs mice
2 laptop_buyers = {101, 102, 103, 105}
3 mouse_buyers = {102, 105, 108, 109}
4
5 # Intersection: Users who bought BOTH
6 both = laptop_buyers.intersection(mouse_buyers)
7 print("Bought both:", both) # {102, 105}
8
9 # Difference: Users who bought laptop but NOT mouse (Prime targets for mouse ads)
10 laptop_only = laptop_buyers.difference(mouse_buyers)
11 print("Recommend mouse to:", laptop_only) # {101, 103}
```

Listing 1.1: Data Science: Set Operations in Python

Derivation: Proving De Morgan's Law

Let's prove the first of De Morgan's Laws: $(A \cup B)^c = A^c \cap B^c$. To prove two sets are equal, we must show that any element in the left set is also in the right set, and vice versa.

1. Let $x \in (A \cup B)^c$.
2. By definition of complement, $x \notin (A \cup B)$.

3. This means x is neither in A nor in B .
4. Logically, this translates to $x \notin A$ **AND** $x \notin B$.
5. Therefore, $x \in A^c$ and $x \in B^c$.
6. By definition of intersection, $x \in (A^c \cap B^c)$.
7. We have proved $(A \cup B)^c \subseteq A^c \cap B^c$. The reverse logic holds symmetrically, proving equality.

Example: Pure Math: Power Set Calculation

Find the power set of $S = \{a, b, c\}$.

- Since $|S| = 3$, the power set must contain $2^3 = 8$ elements.
- The subsets are:

0-element subsets: $\emptyset$

1-element subsets: $\{a\}, \{b\}, \{c\}$

2-element subsets: $\{a, b\}, \{b, c\}, \{a, c\}$

3-element subsets: $\{a, b, c\}$

- Thus, the power set is:

$$\mathcal{P}(S) = \left\{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\} \right\}$$

Example: Data Science: Sets in Recommendation Systems

Let universal set U be the catalog of movies on a streaming platform. Let A be the set of movies watched by User 1, and B be the set of movies watched by User 2.

- $A \cap B$ represents movies both users have watched. If this intersection is large, a recommendation system might infer the users have similar tastes.
- $A \setminus B$ represents movies User 1 has seen but User 2 hasn't. These movies are prime candidates to recommend to User 2!

1.3 Relations

A binary relation R from a set A to a set B is a subset of the Cartesian product $A \times B$. If the ordered pair $(a, b) \in R$, we say " a is related to b " and write aRb .

Core Intuition: Relations as Database Joins

Mathematically, saying a relation is a "subset of a Cartesian product" sounds abstract. Think of it visually like a table in a database:

- Let A be a set of Students, and B be a set of Courses.
- The Cartesian product $A \times B$ generates every possible combination of every student with every course.
- A **Relation** R (e.g., "Enrolled In") simply filters out the nonsense. It picks out only the specific $(Student, Course)$ pairs that actually exist in the real world.

When you perform an `INNER JOIN` in SQL, you are quite literally computing a mathematical relation!

Definition: Properties of Relations on a Set A

A relation $R \subseteq A \times A$ (a relation from set A to itself) is:

- **Reflexive:** if aRa for every $a \in A$. (Every element is related to itself).
- **Symmetric:** if $aRb \implies bRa$ for all $a, b \in A$. (If A is related to B , B is also related to A).
- **Transitive:** if $(aRb \wedge bRc) \implies aRc$ for all $a, b, c \in A$.
- **Equivalence Relation:** if it is simultaneously reflexive, symmetric, and transitive.

Data Science Relevance: Equivalence relations are vital for grouping and clustering data. If you define a relation "has the same label as," this forms an equivalence relation. It partitions the dataset into distinct, non-overlapping groups (equivalence classes) based on labels.

Example: Equivalence Relation Proof

Let $A = \mathbb{Z}$. Define a relation R on $\mathbb{Z}$ by: $xRy \iff x - y$ is divisible by 3. Prove that R is an equivalence relation.

1. **Reflexivity:** For any $x \in \mathbb{Z}$, $x - x = 0 = 3 \times 0$. Since 0 is divisible by 3, xRx is true. Thus, R is reflexive.
2. **Symmetry:** If xRy , then $x - y = 3k$ for some integer k . Multiply by -1 : $y - x = -3k = 3(-k)$. Since $-k$ is also an integer, $y - x$ is divisible by 3. Thus, yRx , and R is symmetric.
3. **Transitivity:** If xRy and yRz , then $x - y = 3k$ and $y - z = 3m$ for some integers k, m . Add the equations: $(x - y) + (y - z) = 3k + 3m \implies x - z = 3(k + m)$. Since $k + m$ is an integer, $x - z$ is divisible by 3. Thus, xRz , and R is transitive.

Since R satisfies all three properties, it is an equivalence relation.

Video Resource: IITM BS Lecture: W1_L6 Relations 

Official IIT Madras lecture covering cartesian products, equivalence, and partitioning sets.

1.4 Functions

A **function** f from set A to set B (denoted $f: A \rightarrow B$) is a special type of relation that associates each element $x \in A$ with **exactly one** element $y \in B$, written as $y = f(x)$.

Data Science Relevance: The entire goal of supervised machine learning is to approximate an unknown mathematical function f that maps input features X to target outputs y ($y = f(X)$). Understanding function properties helps you understand model expressiveness.

Definition: Function Terms and Types

Let $f: A \rightarrow B$.

- **Domain:** The set of all valid inputs (A).
- **Codomain:** The target set containing all possible outputs (B).
- **Range/Image:** The set of actual outputs produced by the function: $\text{Im}(f) = \{f(x) \mid x \in A\}$. Note that the range is a subset of the codomain ($\text{Im}(f) \subseteq B$).
- **One-to-One (Injective):** f maps distinct inputs to distinct outputs. No two inputs share the same output:

$$\forall x_1, x_2 \in A (f(x_1) = f(x_2) \implies x_1 = x_2)$$

- **Onto (Surjective):** The range equals the codomain. Every element in B has at least one pre-image in A :

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y$$

- **Bijection (One-to-One Correspondence):** A function that is both injective and surjective. Bijections are perfectly invertible.

Example: Math & DS: Injectivity and Surjectivity Analysis

Analyze the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x + 5$.

- **Injectivity:** Let $f(x_1) = f(x_2)$.

$$2x_1 + 5 = 2x_2 + 5 \implies 2x_1 = 2x_2 \implies x_1 = x_2$$

Thus, f is injective. In a data context, this means this transformation is perfectly unique; no two distinct data points will be transformed to the same value.

- **Surjectivity:** Let $y \in \mathbb{R}$ (codomain). We want to find an $x \in \mathbb{R}$ (domain) such that $f(x) = y$.

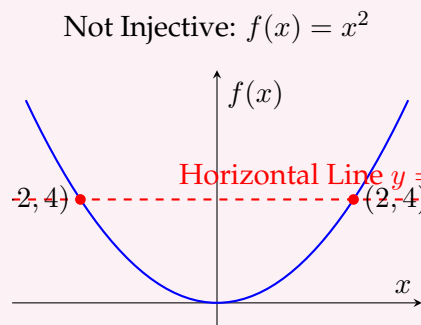
$$2x + 5 = y \implies 2x = y - 5 \implies x = \frac{y - 5}{2}$$

Since $y \in \mathbb{R}$, $x = \frac{y-5}{2}$ is a valid real number in the domain. Checking: $f(x) = 2\left(\frac{y-5}{2}\right) + 5 = (y - 5) + 5 = y$. Thus, f is surjective.

- Since f is both injective and surjective, it is a **bijection**.

Core Intuition: Visualizing Injectivity: The Horizontal Line Test

A quick way to determine if a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is injective is to graph it. Since an injective function can only have one unique x input for any y output, **no horizontal line can intersect the graph more than once**.



Notice how $f(2) = 4$ and $f(-2) = 4$. Two distinct inputs map to the exact same output, violating injectivity!

Video Resource: IITM BS Lecture: W1_L7 Functions



Official IIT Madras lecture specifically covering domain, codomain, range, and properties of mappings.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W1_L0: Introduction to the course basics](#)
- [W1_L1: Natural numbers & their operations](#)
- [W1_L2: Rational numbers - fractions & their properties](#)

- W1_L5: Construction of subsets & set operations
- W1_L5A: Sets examples - membership subsets & comprehension
- W1_L5B: Examples of set operations & counting problems
- W1_L7A: Relations examples - cartesian products tables & join operations
- W1_L7B: Function examples - domains ranges & growth of functions
- W1_L8: Prime numbers - euclids proof infinitude & applications in cryptography
- W1_L9: Why is a number irrational - proof of $\sqrt{2}$
- W1_L10: Sets versus collections - russells paradox & mathematical foundations
- W1_L11: Degrees of infinity - countable & uncountable sets

1.5 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Find $(A \cup B) \setminus (A \cap B)$. This operation is known as the symmetric difference.
2. **Data Science:** A dataset contains a set of users who clicked an ad (C) and users who bought a product (B). Express "Users who clicked the ad but did not buy the product" using set notation.
3. **Pure Math:** Determine if the relation R on $\mathbb{R}$ defined by $xRy \iff x < y$ is an equivalence relation. Explain why or why not.
4. **Data Science:** You define a relation R among users in a database: User A is related to User B if they live in the same city. Is this an equivalence relation?
5. **Pure Math:** Prove whether the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2$ is injective, surjective, both, or neither.
6. **Data Science (Feature Scaling):** A common preprocessing step in data science is Min-Max scaling. Consider a simplified version $g: [0, 10] \rightarrow [0, 1]$ defined by $g(x) = \frac{x}{10}$. Prove that g is a bijection.

Chapter 2

Coordinate Geometry and Straight Lines

Coordinate geometry (or analytic geometry) provides a powerful way to link algebraic equations with geometric shapes. By assigning coordinates to points in a plane, we can analyze geometric properties using algebra. In data science, this forms the foundation of spatial analysis, allowing algorithms to model relationships between features, measure distances between data points, and draw boundaries between different classes.

2.1 Fundamental Formulas

The Cartesian coordinate system represents every point in a two-dimensional plane using a pair of numerical coordinates (x, y) . The horizontal axis is the x-axis, and the vertical axis is the y-axis.

2.1.1 Distance Formula

The straight-line distance d between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is derived from the Pythagorean theorem:

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Data Science Relevance: The distance formula is identical to the **Euclidean Distance**, which is the core metric used by many machine learning algorithms. For example, the K-Nearest Neighbors (KNN) algorithm calculates the distance between a new data point and all existing points to classify it based on its closest neighbors.

```
1 import numpy as np
2
3 # Define two data points (e.g., house features)
4 p = np.array([3, 4])
5 q = np.array([7, 1])
6
7 # Calculate Euclidean Distance
8 distance = np.linalg.norm(p - q)
9 print(f"Distance: {distance}") # Output: 5.0
```

Listing 2.1: Data Science: Euclidean Distance in Python

Video Resource: IITM BS Lecture: W2_L13 Distance Formula



Official IIT Madras lecture calculating the distance between two points in coordinate geometry.

2.1.2 Section Formula

The section formula helps us find the coordinates of a point that divides a line segment into a specific ratio.

Key Formula: Internal and External Division

- **Internal Division:** The point $R(x, y)$ dividing the segment PQ internally in the ratio $m : n$ is:

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}$$

- **Midpoint Formula:** The midpoint of PQ (a 1 : 1 ratio) simplifies to the average of the coordinates:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **External Division:** If division is external:

$$x = \frac{mx_2 - nx_1}{m - n}, \quad y = \frac{my_2 - ny_1}{m - n}$$

Video Resource: IITM BS Lecture: W2_L14 Section Formula



Official IIT Madras lecture explaining how to divide a line segment in a given ratio.

2.2 Slope of a Line

The slope (or gradient) m represents the steepness and direction of a line. For a non-vertical line passing through $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \tan(\theta)$$

where θ is the angle of inclination of the line with the positive x-axis ($0 \leq \theta < \pi, \theta \neq \frac{\pi}{2}$).

Theorem/Rule: Parallel and Perpendicular Lines

Let L_1 and L_2 be two non-vertical lines with slopes m_1 and m_2 respectively.

- **Parallel Condition ($L_1 \parallel L_2$):** The lines never intersect and have the exact same steepness.

$$m_1 = m_2$$

- **Perpendicular Condition ($L_1 \perp L_2$):** The lines intersect at a 90-degree angle.

$$m_1 \cdot m_2 = -1 \iff m_1 = -\frac{1}{m_2}$$

Video Resource: IITM BS Lecture: W2_L16 Slope of a line



Official IIT Madras lecture covering rise over run and the angle of inclination.

2.3 Equations of a Straight Line

Depending on the given parameters, a straight line can be mathematically represented in several different standard forms.

Definition: Standard Forms of Line Equations

- **Slope-Intercept Form:** Given slope m and y-intercept c :

$$y = mx + c$$

- **Point-Slope Form:** Given slope m and a point (x_1, y_1) :

$$y - y_1 = m(x - x_1)$$

- **Two-Point Form:** Passing through (x_1, y_1) and (x_2, y_2) :

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

- **Intercept Form:** Given x-intercept a and y-intercept b :

$$\frac{x}{a} + \frac{y}{b} = 1$$

- **General Form:** Any straight line can be written as:

$$Ax + By + C = 0$$

with slope $m = -A/B$ and y-intercept $c = -C/B$ (if $B \neq 0$).

Example: Finding Line Equations

Find the equation of the line passing through $(-1, 3)$ that is perpendicular to the line $3x - 4y + 8 = 0$.

1. First, rewrite the given line to find its slope:

$$3x - 4y + 8 = 0 \implies 4y = 3x + 8 \implies y = \frac{3}{4}x + 2$$

So, the slope of the given line is $m_1 = \frac{3}{4}$.

2. Since our target line is perpendicular, its slope m_2 satisfies:

$$m_2 = -\frac{1}{m_1} = -\frac{4}{3}$$

3. Now use the point-slope form with slope $m_2 = -\frac{4}{3}$ and point $(-1, 3)$:

$$y - 3 = -\frac{4}{3}(x - (-1)) \implies 3(y - 3) = -4(x + 1)$$

$$3y - 9 = -4x - 4 \implies 4x + 3y - 5 = 0$$

The general equation of the line is $4x + 3y - 5 = 0$.

Video Resource: IITM BS Lecture: W2_L20 General Equation of a Line 

Official IIT Madras lecture discussing forms, conversions, and geometric properties.

2.4 Distance Metrics for Lines

Just as we measure the distance between two points, we can measure the distance from a point to an entire line.

Key Formula: Line Distance Formulas

- **Distance from a Point to a Line:** The perpendicular distance d from point (x_0, y_0) to the line $Ax + By + C = 0$ is:

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

- **Distance between Parallel Lines:** The distance d between two parallel lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$ is:

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

Data Science Relevance: In Support Vector Machines (SVMs), the algorithm tries to find a hyperplane (a line in 2D space) that cleanly separates two classes of data. To do this, it maximizes the perpendicular distance (margin) between the separating line and the closest data points (support vectors). The formula above is exactly what the SVM is optimizing!

2.5 Straight-Line Fit (Least Squares Regression)

In data science, we often want to find a line of best fit $y = mx + c$ for a set of N noisy data points (x_i, y_i) . Under the **ordinary least squares (OLS)** framework, we define "best fit" as the line that minimizes the sum of squared vertical residuals (the error between the actual y values and the predicted values on the line):

$$E(m, c) = \sum_{i=1}^N (y_i - (mx_i + c))^2$$

Theorem/Rule: Least Squares Coefficients

Using calculus (setting the partial derivatives with respect to m and c to zero), we can find the optimal slope and intercept:

$$m = \frac{N \sum (x_i y_i) - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}, \quad c = \bar{y} - m\bar{x}$$

where $\bar{x} = \frac{1}{N} \sum x_i$ and $\bar{y} = \frac{1}{N} \sum y_i$ are the sample means.

Video Resource: IITM BS Lecture: W2_L24 Straight Line Fit 

Official IIT Madras lecture explaining vertical distance, sum of squared errors, and basic regression.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W2_L12: Rectangular coordinate system - axes points lines & quadrants](#)
- [W2_L15: Area of a triangle - formula derivation using coordinates](#)
- [W2_L17: Parallel & perpendicular lines - slope conditions & angle of intersection](#)
- [W2_L18: Representation of a line 1 - point slope & two point forms](#)
- [W2_L19: Representation of a line 2 - slope intercept & intercept forms](#)
- [W2_L21: Equation of parallel & perpendicular lines in general form](#)
- [W2_L22: Equation of a perpendicular line through a point using slope conditions](#)
- [W2_L23: Distance of a point from a line - perpendicular distance & parallel lines](#)

2.6 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Find the distance between the points $A(3, 4)$ and $B(7, 1)$.
2. **Pure Math:** A point P divides the line segment joining $X(1, -2)$ and $Y(-3, 6)$ internally in the ratio $1 : 3$. Find the coordinates of P .
3. **Data Science:** A K-Means clustering algorithm places a centroid at $(5, 5)$. A new data point arrives at $(2, 9)$. What is the Euclidean distance from the point to the centroid?
4. **Pure Math:** Find the equation of the line passing through the points $(2, 3)$ and $(4, 7)$ in slope-intercept form.
5. **Data Science:** A simple linear regression model predicts housing prices (y) based on square footage (x). The model parameters are learned as slope $m = 150$ and intercept $c = 50000$. If a house has 2000 square feet, what is the predicted price?
6. **Pure Math:** Find the perpendicular distance from the origin $(0, 0)$ to the line $3x - 4y + 15 = 0$.

Chapter 3

Quadratic Functions

Quadratic functions are second-degree polynomials. Geometrically, they represent parabolas. In data science, quadratic functions are widely used in loss functions (like mean squared error) and optimization problems, as they possess a single local (and global) extremum.

3.1 Quadratic Equations and Solving Methods

A quadratic equation is of the form:

$$ax^2 + bx + c = 0 \quad (\text{where } a, b, c \in \mathbb{R} \text{ and } a \neq 0)$$

Key Formula: Solving Quadratic Equations

We can find the roots (solutions) using three main approaches:

1. **Factoring:** Rewriting the equation as $a(x - r_1)(x - r_2) = 0$.
2. **Completing the Square:** Expressing the equation in the form $(x - h)^2 = k$.
3. **The Quadratic Formula:**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Data Science Relevance: Many optimization algorithms, such as Gradient Descent, are designed to find the roots or extrema of functions. While quadratics can be solved analytically using the formula above, more complex functions require iterative numeric solvers.

Video Resource: IITM BS Lecture: W3_L3A Solution by Factorization 

Official IIT Madras lecture covering the factorization method for quadratic roots.

3.1.1 Nature of Roots (The Discriminant)

The term under the square root in the quadratic formula is the **discriminant**, denoted by D (or Δ):

$$D = b^2 - 4ac$$

Theorem/Rule: Discriminant Properties

For a quadratic equation with real coefficients:

- If $D > 0$: The equation has **two distinct real roots**. The parabola crosses the x-axis twice.
- If $D = 0$: The equation has **one repeated real root** at $x = -\frac{b}{2a}$. The parabola just touches the x-axis.
- If $D < 0$: The equation has **two complex conjugate roots** ($x = \frac{-b \pm i\sqrt{|D|}}{2a}$). The graph does not touch the x-axis.

Video Resource: IITM BS Lecture: W3_L4 Quadratic Formula & Discriminant



Official IIT Madras lecture discussing completing the square, the discriminant, and the nature of roots.

3.2 Geometry of the Parabola

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is a **parabola**.

Definition: Parabolic Characteristics

- **Direction of Opening:**
 - If $a > 0$, the parabola opens **upward** (convex).
 - If $a < 0$, the parabola opens **downward** (concave).
- **Vertex (V):** The turning point of the parabola. The vertex coordinates are:

$$V(h, k) = \left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right) \right) = \left(-\frac{b}{2a}, -\frac{b^2 - 4ac}{4a} \right)$$

- **Vertex Form:** Any quadratic function can be written as:

$$f(x) = a(x - h)^2 + k$$

where (h, k) is the vertex.

- **Axis of Symmetry:** The vertical line passing through the vertex:

$$x = -\frac{b}{2a}$$

Video Resource: IITM BS Lecture: W4_L1A Quadratic Functions & Parabolas



Official IIT Madras lecture analyzing the vertex, graphs, and overall shape of a parabola.

Example: Completing the Square and Graphing

Convert the function $f(x) = 2x^2 - 8x + 6$ into vertex form, find its vertex, axis of symmetry, and intercepts.

1. Factor out the coefficient $a = 2$ from the variable terms:

$$f(x) = 2(x^2 - 4x) + 6$$

2. Complete the square inside the parenthesis. Half of -4 is -2 , and $(-2)^2 = 4$. Add and subtract 4:

$$f(x) = 2(x^2 - 4x + 4 - 4) + 6 = 2((x - 2)^2 - 4) + 6$$

$$f(x) = 2(x - 2)^2 - 8 + 6 \implies f(x) = 2(x - 2)^2 - 2$$

3. **Vertex (h, k) :** The vertex is $(2, -2)$. Since $a = 2 > 0$, it is a minimum point.

4. **Axis of Symmetry:** The vertical line $x = 2$.

5. **Intercepts:**

- y-intercept: $f(0) = 2(0)^2 - 8(0) + 6 = 6 \implies (0, 6)$.
- x-intercepts: Set $f(x) = 0 \implies 2(x - 2)^2 - 2 = 0 \implies 2(x - 2)^2 = 2 \implies (x - 2)^2 = 1 \implies x - 2 = \pm 1$. So $x = 3$ or $x = 1$. Thus, intercepts are $(1, 0)$ and $(3, 0)$.

3.3 Optimization Applications

Because quadratics have a single extremum at their vertex, optimization is straightforward. This is why loss functions like Mean Squared Error (MSE) are formulated quadratically: they guarantee a single, easily discoverable minimum.

Theorem/Rule: Quadratic Extremum

For the function $f(x) = ax^2 + bx + c$:

- If $a > 0$, $f(x)$ achieves its **global minimum** value of $-\frac{D}{4a}$ at $x = -\frac{b}{2a}$.
- If $a < 0$, $f(x)$ achieves its **global maximum** value of $-\frac{D}{4a}$ at $x = -\frac{b}{2a}$.

Example: Business Revenue Optimization

A company's daily profit is modeled by $P(x) = -10x^2 + 1200x - 2000$, where x is the price of their product in dollars. Find the price that maximizes profit, and calculate that maximum profit.

- Here, $a = -10 < 0$ and $b = 1200$. Since $a < 0$, the profit function is a downward-opening parabola, meaning it has a maximum at its vertex.
- The maximizing price x is:

$$x = -\frac{b}{2a} = -\frac{1200}{2(-10)} = \frac{1200}{20} = 60 \text{ dollars}$$

- The maximum profit is $P(60)$:

$$P(60) = -10(60)^2 + 1200(60) - 2000 = -10(3600) + 72000 - 2000$$

$$P(60) = -36000 + 72000 - 2000 = 34000 \text{ dollars}$$

So, setting the price to \$60 maximizes daily profit at \$34,000.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W3_Part B: Surface of revolution | cone formation by rotating a line](#)
- [W3_L2B: Summary lecture](#)
- [W3_L3B: Solution of a quadratic equation using square method](#)
- [W3_Part A: Additional lecture | standard form & vertex form of parabola](#)
- [W3_L4A: Summary](#)
- [W4_L1B: Examples of quadratic functions](#)
- [W4_L1C: Slope of a quadratic function](#)
- [W4_L2: Solution of a quadratic equation using graph](#)
- [W4_L2A: Slope line & parabola](#)

3.4 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Find the roots of the quadratic equation $3x^2 - 5x - 2 = 0$ using factorization.
2. **Pure Math:** Calculate the discriminant of $4x^2 + 12x + 9 = 0$ and state the nature of its roots.
3. **Data Science:** A Mean Squared Error loss function for a single parameter w is defined as $L(w) = 2w^2 - 16w + 35$. Find the value of w that minimizes the loss, and the minimum loss value itself.
4. **Pure Math:** Convert the quadratic function $y = x^2 + 6x + 5$ into vertex form and identify its vertex.
5. **Data Science:** A manufacturing algorithm predicts that producing n items yields a cost $C(n) = 0.5n^2 - 50n + 1500$. How many items should be produced to minimize the cost?

Chapter 4

Algebra of Polynomials

A polynomial is a mathematical expression consisting of variables and coefficients, involving only addition, subtraction, multiplication, and non-negative integer exponents. Understanding polynomial algebra is critical for advanced regression techniques and kernel methods in machine learning, which use high-degree polynomials to capture complex, non-linear patterns in data.

4.1 Definitions and Classifications

A real polynomial $P(x)$ of degree n is defined as:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $a_0, a_1, \dots, a_n \in \mathbb{R}$ with $a_n \neq 0$, and $n \in \{0, 1, 2, \dots\}$.

- **Degree (n):** The highest exponent of the variable x with a non-zero coefficient.
- **Leading Coefficient (a_n):** The coefficient of the term with the highest degree.
- **Constant Term (a_0):** The term of degree 0.

Core Intuition: Polynomials as Lego Blocks

Why do mathematicians care so much about x^2 , x^3 , and x^4 ? Think of these simple powers of x as the "Lego blocks" of mathematics. By stacking them together and adjusting their coefficients (a_i), you can bend a curve to mimic almost any shape in the real world!

- Need a straight ramp? Use $a_1 x$.
- Need a parabolic satellite dish? Add $a_2 x^2$.
- Need an S-curve that models population growth? Add $a_3 x^3$.

In calculus, you will later learn the **Taylor Series**, which proves that *any* smooth, complicated function (like sine waves or logarithms) can be approximated perfectly just by stacking enough polynomial Lego blocks!

Data Science Relevance: In **Polynomial Regression**, if a straight line (degree 1) doesn't fit the data well, we might fit a quadratic (degree 2) or cubic (degree 3) polynomial. However, choosing too high of a degree leads to "overfitting" the model perfectly learns the training data but fails to generalize to new data.

Video Resource: IITM BS Lecture: W4_L1 Introduction to Polynomials 

Official IIT Madras lecture covering the definition and types of polynomials.

4.2 Polynomial Division and Theorems

Like integers, polynomials can be divided.

Theorem/Rule: Division Algorithm for Polynomials

For any dividend polynomial $P(x)$ and non-zero divisor polynomial $D(x)$, there exist unique polynomials $Q(x)$ (quotient) and $R(x)$ (remainder) such that:

$$P(x) = D(x)Q(x) + R(x)$$

where either $R(x) = 0$ or $\deg(R) < \deg(D)$.

Theorem/Rule: Remainder and Factor Theorems

Let $P(x)$ be a polynomial of degree $n \geq 1$ and let $c \in \mathbb{R}$.

- **Remainder Theorem:** If $P(x)$ is divided by the linear term $(x - c)$, the remainder is exactly the value $P(c)$.
- **Factor Theorem:** The linear term $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$.

Example: Remainder Theorem and Factoring

Determine if $(x - 2)$ is a factor of $P(x) = x^3 - 3x^2 + 4x - 4$, and find the remainder when divided.

- By the Remainder Theorem, the remainder R is equal to $P(2)$:

$$P(2) = (2)^3 - 3(2)^2 + 4(2) - 4 = 8 - 3(4) + 8 - 4 = 8 - 12 + 8 - 4 = 0$$

- Since $P(2) = 0$, the remainder is 0.
- By the Factor Theorem, because the remainder is 0, $(x - 2)$ **is a factor** of $P(x)$.

Video Resource: IITM BS Lecture: W4_L6 Polynomial Division Algorithm 

Official IIT Madras lecture explaining long division with remainders.

4.3 Graphing Polynomials

To sketch a polynomial function $P(x) = a_n x^n + \cdots + a_0$, we analyze three key features: End Behavior, Intercepts/Multiplicities, and Turning Points.

4.3.1 End Behavior

The long-term behavior of $P(x)$ as $x \rightarrow \pm\infty$ is determined solely by its leading term $a_n x^n$.

Theorem/Rule: Leading Coefficient Test for End Behavior

	Even Degree (n is even)	Odd Degree (n is odd)
Positive a_n	$x \rightarrow \infty, y \rightarrow \infty$ $x \rightarrow -\infty, y \rightarrow \infty$ (Up-Up)	$x \rightarrow \infty, y \rightarrow \infty$ $x \rightarrow -\infty, y \rightarrow -\infty$ (Down-Up)
Negative a_n	$x \rightarrow \infty, y \rightarrow -\infty$ $x \rightarrow -\infty, y \rightarrow -\infty$ (Down-Down)	$x \rightarrow \infty, y \rightarrow -\infty$ $x \rightarrow -\infty, y \rightarrow \infty$ (Up-Down)

Video Resource: IITM BS Lecture: W4_L9 End Behavior of Polynomials



Official IIT Madras lecture explaining leading terms, degrees, and the asymptotic limits of polynomials.

4.3.2 Intercepts and Root Multiplicities

If $P(c) = 0$, then c is a real zero (root) of $P(x)$ and $(c, 0)$ is an x-intercept. If a factor $(x - c)$ appears repeated k times (i.e., $(x - c)^k$ divides $P(x)$), then c is a zero of **multiplicity** k .

Definition: Multiplicity Behavior at X-intercepts

- **Odd Multiplicity (k is odd):** The graph **crosses** the x-axis at $x = c$. If $k > 1$, the graph also flattens (inflection point) as it crosses.
- **Even Multiplicity (k is even):** The graph is **tangent** to the x-axis (touches and bounces/turns back) at $x = c$.

4.3.3 Turning Points

A turning point is a point where the graph changes direction from increasing to decreasing, or vice versa (local extrema).

Theorem/Rule: Turning Points Bound

A polynomial of degree n can have **at most** $n - 1$ turning points.

Example: Polynomial Sketching Analysis

Analyze the polynomial $f(x) = -(x - 1)^2(x + 2)$ for sketching.

1. **Degree and Leading Coefficient:** Expanding the leading term: $f(x) = -(x^2)(x) + \dots = -x^3 + \dots$. The degree is $n = 3$ (odd), and the leading coefficient is $a_n = -1$ (negative). End behavior: As $x \rightarrow \infty, y \rightarrow -\infty$. As $x \rightarrow -\infty, y \rightarrow \infty$ (Up-Down).
2. **Intercepts:**
 - y-intercept: $f(0) = -(0 - 1)^2(0 + 2) = -1(1)(2) = -2 \implies (0, -2)$.
 - x-intercepts: Set $f(x) = 0 \implies (x - 1)^2(x + 2) = 0$. Zeros: $x = 1$ (multiplicity 2) and $x = -2$ (multiplicity 1).
3. **Behavior at Zeros:**
 - At $x = 1$ (even multiplicity): The graph touches and bounces off the x-axis.
 - At $x = -2$ (odd multiplicity): The graph crosses the x-axis.
4. **Turning Points:** At most $n - 1 = 2$ turning points.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W4_L2: Degree of polynomials definition examples & special cases](#)
- [W4_L3: Algebra of polynomials addition & subtraction](#)
- [W4_L4: Algebra of polynomials multiplication methods](#)
- [W4_L5: Algebra of polynomials division with remainders](#)
- [W4_L7: Graphs of polynomials identification & characterization](#)
- [W4_L8: Graphs of polynomials behavior at x intercepts & multiplicities](#)
- [W4_L10: Graphs of polynomials turning points & relation to degree](#)
- [W4_L11: Graphs of polynomials graphing methods & polynomial creation](#)
- [W6: Weekly summary: polynomial definitions & algebra operations](#)
- [W7: Summary: polynomial graphs & real-world applications](#)

4.4 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Use the Remainder Theorem to find the remainder when $P(x) = x^4 - 2x^3 + 5x - 7$ is divided by $(x - 1)$.
2. **Pure Math:** Determine the end behavior of the polynomial $f(x) = 3x^5 - 4x^3 + 2x - 1$.
3. **Data Science:** You are fitting a polynomial regression curve to 4 data points. What is the maximum number of turning points a degree-3 polynomial curve can have as it passes through the data?
4. **Pure Math:** Find the x-intercepts and state their multiplicities for the polynomial $g(x) = x^2(x + 3)^3(x - 5)$. Will the graph cross or bounce at $x = -3$?
5. **Data Science:** A complex machine learning boundary is defined by the polynomial equation $B(x) = -2x^6 + 5x^4 - 3x^2 + 1$. As $x \rightarrow \infty$ and $x \rightarrow -\infty$, in which direction does the boundary eventually head?

Chapter 5

Functions, Compositions, and Exponentials

This week deepens our understanding of functions, focusing on how functions can be combined (composites), reversed (inverses), and exponentially scaled.

5.1 Testing Functions: Vertical and Horizontal Lines

Graphs are an excellent way to visually analyze the properties of functions and mappings. We use two geometric tests:

Theorem/Rule: Line Tests for Functions

- **Vertical Line Test (Function Existence):** A curve in the Cartesian plane represents a valid function $y = f(x)$ if and only if **no vertical line** intersects the curve at more than one point. If a vertical line hits twice, one input x is mapping to multiple outputs, which violates the definition of a function.
- **Horizontal Line Test (One-to-One / Injectivity):** A function $y = f(x)$ is one-to-one (injective) if and only if **no horizontal line** intersects its graph at more than one point. If a horizontal line hits twice, multiple inputs map to the same output.

Video Resource: IITM BS Lecture: W5_L1 One-to-One Function Tests



Official IIT Madras lecture covering definitions and tests for injectivity.

5.2 Composite Functions

Composition is the process of applying one function to the results of another.

Definition: Composite Function

Let $g: A \rightarrow B$ and $f: B \rightarrow C$ be functions. The **composite function** $f \circ g$ (read "f of g") is a function from A to C defined by:

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ lies in the domain of f :

$$\text{Domain}(f \circ g) = \{x \in \text{Domain}(g) \mid g(x) \in \text{Domain}(f)\}$$

Data Science Relevance: Artificial Neural Networks (Deep Learning) are fundamentally massive chains of composite functions. The output of one layer of neurons ($g(x)$) becomes the direct input for the next layer ($f(g(x))$). By chaining simple functions together, networks can approximate incredibly complex boundaries.

Example: Function Composition and Domain

Let $f(x) = \sqrt{x}$ and $g(x) = x - 3$. Find $(f \circ g)(x)$ and its domain.

- Find the expression:

$$(f \circ g)(x) = f(g(x)) = f(x - 3) = \sqrt{x - 3}$$

- Find the domain: The domain of $g(x)$ is all real numbers $\mathbb{R}$. The domain of $f(t) = \sqrt{t}$ requires $t \geq 0$. Therefore, we require $g(x) \geq 0 \implies x - 3 \geq 0 \implies x \geq 3$. Thus, the domain of $f \circ g$ is $[3, \infty)$.

Video Resource: IITM BS Lecture: W5_L6 Composite Functions



Official IIT Madras lecture covering the definitions, domains, and real-world examples of composites.

5.3 Inverse Functions

An inverse function reverses the mapping of the original function.

Definition: Inverse Function

If $f: A \rightarrow B$ is a **bijection** (both one-to-one and onto), then there exists a unique function $f^{-1}: B \rightarrow A$ called the **inverse of f** such that:

$$f^{-1}(y) = x \iff f(x) = y \quad \forall y \in B$$

Key properties:

- $\text{Domain}(f^{-1}) = \text{Range}(f)$ and $\text{Range}(f^{-1}) = \text{Domain}(f)$.
- $f(f^{-1}(y)) = y$ and $f^{-1}(f(x)) = x$.
- The graph of f^{-1} is the reflection of the graph of f across the diagonal line $y = x$.

Example: Finding an Inverse Function

Find the inverse of the function $f(x) = \frac{2x-1}{x+3}$ defined for $x \neq -3$.

1. Write $y = f(x)$:

$$y = \frac{2x-1}{x+3}$$

2. Solve for x in terms of y :

$$y(x+3) = 2x-1 \implies xy+3y = 2x-1$$

Collect all terms with x :

$$xy - 2x = -3y - 1 \implies x(y-2) = -(3y+1) \implies x = \frac{3y+1}{2-y}$$

3. Swap the variables to write $f^{-1}(x)$:

$$f^{-1}(x) = \frac{3x+1}{2-x} \quad (\text{defined for } x \neq 2)$$

Video Resource: IITM BS Lecture: W5_L9 Inverse Functions



Official IIT Madras lecture covering definitions and graphical symmetry of inverse functions.

5.4 Exponential Functions

An exponential function is a function where the variable is placed in the exponent, leading to extremely rapid growth (or decay).

Definition: Exponential Function

An exponential function is of the form:

$$f(x) = a^x \quad (\text{where } a > 0 \text{ and } a \neq 1)$$

- **Domain:** $(-\infty, \infty)$ (all real numbers).
- **Range:** $(0, \infty)$ (always strictly positive, never 0).
- **Natural Exponential Function:** When the base is Euler's number $e \approx 2.71828$:

$$f(x) = e^x$$

Theorem/Rule: Laws of Exponents

For $a > 0$ and any real numbers x, y :

- $a^x \cdot a^y = a^{x+y}$
- $\frac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{xy}$
- $(ab)^x = a^x b^x$

Data Science Relevance: The natural exponential e^x is pervasive in probability, statistics, and machine learning (especially Logistic Regression and the Softmax function) because its rate of growth is exactly equal to its current value a unique and mathematically elegant property.

Video Resource: IITM BS Lecture: W5_L3 Exponential Functions Definition 

Official IIT Madras lecture introducing exponential properties and growth.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W5_L2: One-to-one function examples, cubic cases & theorems](#)
- [W5_L4: Exponential functions graphing with examples & analysis](#)
- [W5_L5: Natural exponential function definition & properties](#)
- [W5_L7: Composite functions step by step examples](#)
- [W5_L8: Composite functions finding the domain](#)
- [W8: Additional lecture: inverse functions finding & verifying](#)

5.5 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Use the horizontal line test conceptually to determine if $f(x) = x^2$ is one-to-one on the domain $\mathbb{R}$. What if the domain is restricted to $[0, \infty)$?
2. **Pure Math:** Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Are they equal?
3. **Data Science:** In a neural network, a linear transformation $L(x) = 3x - 2$ is followed by a ReLU activation function $R(x) = \max(0, x)$. Evaluate the composite output $(R \circ L)(1)$ and $(R \circ L)(0)$.
4. **Pure Math:** Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{4x}{x-2}$.
5. **Pure Math:** Simplify the exponential expression: $\frac{(e^{2x})^3 \cdot e^{-x}}{e^{4x}}$.

Chapter 6

Logarithmic Functions

Logarithms are the mathematical inverses of exponential operations. While exponentials represent rapid compound growth, logarithms scale that growth back down, effectively compressing large spans of numbers into manageable ranges. They are fundamental in data science for creating metrics like entropy, cross-entropy loss, and for stabilizing computationally large datasets.

6.1 Definition of Logarithmic Functions

Definition: Logarithmic Function

For $a > 0$ and $a \neq 1$, the logarithmic function with base a , denoted $\log_a(x)$, is defined as the inverse of the exponential function $y = a^x$:

$$y = \log_a(x) \iff a^y = x \quad (\text{for } x > 0)$$

- **Domain:** $(0, \infty)$ (you cannot take the log of a negative number or zero).
- **Range:** $(-\infty, \infty)$ (can output any real number).
- **Vertical Asymptote:** The line $x = 0$ (y-axis) is a vertical asymptote because as $x \rightarrow 0^+$, $y \rightarrow -\infty$ (for $a > 1$).
- **Natural Logarithm:** The log with base e :

$$\ln(x) = \log_e(x)$$

- **Common Logarithm:** The log with base 10:

$$\log(x) = \log_{10}(x)$$

Data Science Relevance: In data preprocessing, if a dataset has a heavily right-skewed distribution (like income data where a few billionaires warp the scale), applying a **Log Transformation** ($y = \log(x)$) compresses the massive outliers and expands the smaller values, often resulting in a more normal, bell-shaped distribution that machine learning models prefer.

```
1 import numpy as np
2
3 # Highly skewed income data
4 incomes = np.array([30000, 45000, 1000000, 50000, 2500000])
5
6 # Apply Natural Log transformation to compress outliers
```

```

7 log_incomes = np.log(incomes)
8 print("Original:", incomes)
9 print("Log transformed:", np.round(log_incomes, 2))

```

Listing 6.1: Data Science: Log Transformation in Python

Video Resource: IITM BS Lecture: W6_L1 Logarithms & Inverses 

Official IIT Madras lecture defining logarithmic functions as the direct inverse of exponentials.

6.2 Laws of Logarithms

Because logarithms are essentially exponents, they follow algebraic rules derived directly from the laws of exponents. These rules are computationally powerful because they turn multiplication into addition, which computers can process much faster and with less risk of numerical overflow.

Theorem/Rule: Laws of Logarithms

For any positive numbers x, y , base $a > 0, a \neq 1$, and real number p :

- **Product Rule:** $\log_a(xy) = \log_a(x) + \log_a(y)$
- **Quotient Rule:** $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
- **Power Rule:** $\log_a(x^p) = p \log_a(x)$
- **Base Identity:** $\log_a(a) = 1$ and $\log_a(1) = 0$
- **Inverse Properties:** $\log_a(a^x) = x$ and $a^{\log_a(x)} = x$
- **Change of Base Formula:** For any new positive base b :

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)} = \frac{\ln(x)}{\ln(a)}$$

Example: Simplifying Logarithmic Expressions

Simplify the expression: $3 \ln(x) + \frac{1}{2} \ln(y) - 2 \ln(z)$ into a single logarithm.

1. Apply the power rule first to move the coefficients into the exponents:

$$\ln(x^3) + \ln(y^{1/2}) - \ln(z^2)$$

2. Apply the product rule to combine the first two terms being added:

$$\ln(x^3 \cdot \sqrt{y}) - \ln(z^2)$$

3. Apply the quotient rule to incorporate the subtracted term:

$$\ln\left(\frac{x^3 \sqrt{y}}{z^2}\right)$$

Video Resource: IITM BS Lecture: W6_L4 Logarithmic Laws & Properties 

Official IIT Madras lecture proving and demonstrating the fundamental laws of logarithms.

6.3 Solving Equations

We use the inverse relationship between exponentials and logarithms to isolate variables that are trapped inside exponents or log functions.

6.3.1 Solving Exponential Equations

When the variable is in the exponent, we take the logarithm of both sides to bring the variable down.

Example: Solving Exponential Equations

Solve the equation: $5^{2x-1} = 3^x$.

1. Take the natural logarithm ($\ln$) of both sides:

$$\ln(5^{2x-1}) = \ln(3^x)$$

2. Apply the power rule to bring down the exponents:

$$(2x - 1) \ln(5) = x \ln(3)$$

3. Expand the left side:

$$2x \ln(5) - \ln(5) = x \ln(3)$$

4. Gather all terms containing x on one side:

$$2x \ln(5) - x \ln(3) = \ln(5)$$

5. Factor out x :

$$x(2 \ln(5) - \ln(3)) = \ln(5)$$

6. Solve for x :

$$x = \frac{\ln(5)}{2 \ln(5) - \ln(3)} = \frac{\ln(5)}{\ln(25) - \ln(3)} = \frac{\ln(5)}{\ln(25/3)}$$

Video Resource: IITM BS Lecture: W6_L3 Solving Exponential Equations 

Official IIT Madras lecture showing how to use logarithms to solve complex exponential formulas.

6.3.2 Solving Logarithmic Equations

When the variable is inside a logarithm, we isolate the logarithm and then rewrite the equation in its exponential form.

Example: Solving Logarithmic Equations

Solve the equation: $\log_2(x) + \log_2(x - 2) = 3$.

1. Combine the logarithms using the product rule:

$$\log_2(x(x - 2)) = 3 \implies \log_2(x^2 - 2x) = 3$$

2. Convert to exponential form to eliminate the log:

$$x^2 - 2x = 2^3 \implies x^2 - 2x = 8$$

3. Move terms to one side to form a standard quadratic equation:

$$x^2 - 2x - 8 = 0 \implies (x - 4)(x + 2) = 0$$

So the potential solutions are $x = 4$ or $x = -2$.

4. **Check for extraneous solutions (Crucial!):**

- If $x = 4$: The original terms $\log_2(4)$ and $\log_2(2)$ are both evaluating positive numbers, which is valid.
- If $x = -2$: The original term $\log_2(-2)$ is undefined because the domain of logs is strictly positive. So $x = -2$ is an extraneous solution.

Thus, the only valid solution is $x = 4$.

Video Resource: IITM BS Lecture: W6_L7 Solving Logarithmic Equations



Official IIT Madras lecture applying properties to solve logarithmic equations and checking domains.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W6_L2: Graphs & properties of logarithmic functions](#)
- [W6_L5: Logarithmic functions applications of log laws](#)
- [W6_L6: Logarithmic functions change of base theorem](#)

6.4 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Evaluate the expression without a calculator: $\log_3(81) - \log_2\left(\frac{1}{8}\right)$.
2. **Data Science:** In binary classification, the Cross-Entropy loss for a true label $y = 1$ and predicted probability p is defined as $L = -\ln(p)$. If the predicted probability is $p = e^{-2}$, what is the loss?
3. **Pure Math:** Use the laws of logarithms to expand the expression completely: $\ln\left(\frac{x^4\sqrt{y^3}}{e^2}\right)$.
4. **Data Science:** A dataset feature X ranges from 1 to 1,000,000. A data scientist applies a base-10 log transformation to create a new feature $X' = \log_{10}(X)$. What is the new range of X' ?
5. **Pure Math:** Solve the equation for x : $e^{2x} - 3e^x + 2 = 0$. (Hint: treat it as a quadratic in form).
6. **Pure Math:** Solve the logarithmic equation: $\log(x + 3) - \log(x) = 1$.

Chapter 7

Sequences and Limits (Part 1)

Calculus is the mathematical study of continuous change. It is entirely built upon the foundational concept of a **limit**. In this chapter, we explore how sequence terms behave as they approach infinity, and how this relates to geometric tangents and instantaneous rates of change.

7.1 Sequences and Convergence

A **sequence** is an ordered list of numbers. Formally, it can be viewed as a function whose domain is the set of natural numbers $\mathbb{N}$, written as:

$$a_1, a_2, a_3, \dots, a_n, \dots \quad \text{OR} \quad \{a_n\}_{n=1}^{\infty}$$

where a_n represents the n -th term of the sequence.

Definition: Limit of a Sequence

A sequence $\{a_n\}$ has the limit L (written as $\lim_{n \rightarrow \infty} a_n = L$) if the terms a_n get arbitrarily close to L as n grows to infinity.

- **Convergent:** If the limit exists and is a finite real number L .
- **Divergent:** If the limit does not exist (e.g., the terms grow without bound toward $\pm\infty$, or they oscillate back and forth indefinitely).

Data Science Relevance: In machine learning, training an algorithm like a neural network involves minimizing a loss function over multiple iterations (epochs). The sequence of loss values $\{L_1, L_2, L_3, \dots, L_n\}$ as training progresses must ideally **converge** to a minimum value. If the loss sequence *diverges*, the model has failed to learn (often due to a learning rate that is too high).

Theorem/Rule: Limit Laws for Sequences

If $\{a_n\}$ and $\{b_n\}$ are convergent sequences with $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$:

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = A \pm B$
- $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$
- $\lim_{n \rightarrow \infty} \left(\frac{a_n}{b_n} \right) = \frac{A}{B}$ (provided $b_n \neq 0$ and $B \neq 0$)
- $\lim_{n \rightarrow \infty} c = c$ (for any constant c)

Example: Sequence Convergence Check

Determine if the sequence $a_n = \frac{4n^2-3n}{2n^2+5n+1}$ converges, and find its limit.

- Divide both the numerator and denominator by the highest power of n (which is n^2):

$$a_n = \frac{\frac{4n^2}{n^2} - \frac{3n}{n^2}}{\frac{2n^2}{n^2} + \frac{5n}{n^2} + \frac{1}{n^2}} = \frac{4 - \frac{3}{n}}{2 + \frac{5}{n} + \frac{1}{n^2}}$$

- Take the limit as $n \rightarrow \infty$. Note that as n grows infinitely large, any fraction with n in the denominator shrinks to 0 ($\lim_{n \rightarrow \infty} \frac{1}{n} = 0$):

$$\lim_{n \rightarrow \infty} a_n = \frac{4 - 0}{2 + 0 + 0} = \frac{4}{2} = 2$$

- Since the limit is a finite number, the sequence **converges** to 2.

Video Resource: Khan Academy: Introduction to Limits



A highly intuitive, visual introduction to what limits actually mean conceptually.

7.2 Rates of Change and Secant Lines

Let $y = f(x)$ be a function representing some changing quantity (e.g., position over time).

Definition: Average Rate of Change and Secants

- The **average rate of change** of $f(x)$ over the interval $[x_1, x_2]$ is the ratio of change in output to change in input:

$$\text{Average Rate of Change} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

- Geometrically, this ratio is the slope of the **secant line** that connects the two points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ on the graph of f .

7.3 Tangents as Limits of Secants

If we want to find the *instantaneous* rate of change at exactly one point $x_1 = c$, we can't use the secant formula directly because $\frac{f(c)-f(c)}{c-c} = \frac{0}{0}$, which is undefined.

Instead, we let the second point x_2 get closer and closer to c . Let the difference between them be h (so $x_2 = c + h$). As $h \rightarrow 0$, the secant line approaches the **tangent line** at c .

Definition: Tangent Line Slope (The Derivative)

The slope of the tangent line to the curve $y = f(x)$ at the point $(c, f(c))$ is the limit of the secant slopes as $h \rightarrow 0$:

$$m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

This instantaneous slope is the foundation of the **derivative**.

Data Science Relevance: In Gradient Descent, we want to slide down a loss curve to find its bottom. To do this, we need to know the slope of the tangent line at our current position. The limit formula above is exactly how that slope is mathematically defined!

Example: Finding Tangent Slope using Limits

Find the slope of the tangent line to the curve $f(x) = x^2 + 2x$ at $x = 3$.

1. Set up the limit for the slope at $c = 3$:

$$m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$$

2. Compute the output at our target point: $f(3) = 3^2 + 2(3) = 9 + 6 = 15$.
3. Compute the output slightly offset by h :

$$f(3+h) = (3+h)^2 + 2(3+h) = (9 + 6h + h^2) + (6 + 2h) = h^2 + 8h + 15$$

4. Substitute back into the limit:

$$m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{(h^2 + 8h + 15) - 15}{h} = \lim_{h \rightarrow 0} \frac{h^2 + 8h}{h}$$

5. Factor out h to resolve the division by zero issue:

$$m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{h(h+8)}{h} = \lim_{h \rightarrow 0} (h+8) = 8$$

So, the instantaneous slope of the tangent line to $x^2 + 2x$ at exactly $x = 3$ is 8.

Video Resource: 3Blue1Brown: The Essence of Calculus



The absolute best visual explanation of how shrinking secant lines transform into tangent derivatives.

7.4 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Determine if the sequence $b_n = \frac{5n^3 - n + 2}{7n^3 + 4n^2}$ converges, and if so, find its limit.
2. **Pure Math:** Determine if the sequence $c_n = \frac{n^2}{3n+1}$ converges or diverges.
3. **Data Science:** A machine learning model's loss over n epochs is modeled by the sequence $L_n = 2 + \frac{10}{n^2}$. As training goes on infinitely ($n \rightarrow \infty$), what is the theoretical minimum loss the model converges to?
4. **Pure Math:** Find the average rate of change of the function $f(x) = x^3$ over the interval $[1, 3]$.
5. **Pure Math:** Using the limit definition of the tangent slope $m_{\tan} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$, find the instantaneous slope of $f(x) = x^2$ at the point $x = 4$.

Chapter 8

Differentiation

Differentiation formalizes the concept of the instantaneous rate of change of a function. In data science, differentiation is the mathematical engine behind optimization algorithms. For example, neural networks use a technique called "backpropagation" to learn, which is fundamentally just an application of the derivative chain rule across millions of parameters.

8.1 Limits, Continuity, and Differentiability

Let $f(x)$ be a function defined on an open interval containing c .

Definition: Limit, Continuity, and Derivative

- **Limit:** We say $\lim_{x \rightarrow c} f(x) = L$ if we can make the values of $f(x)$ arbitrarily close to L by taking x sufficiently close to c . This requires the left-hand limit and right-hand limit to be equal:

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$$

- **Continuity:** $f(x)$ is continuous at $x = c$ if the limit exists and equals the function's value:

$$\lim_{x \rightarrow c} f(x) = f(c)$$

- **Derivative:** The derivative of $f(x)$ at x , denoted $f'(x)$ or $\frac{dy}{dx}$, is defined as the limit of the difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

If this limit exists, f is **differentiable** at x . Differentiability strictly implies continuity, but continuity does **not** guarantee differentiability (e.g., $f(x) = |x|$ is continuous at $x = 0$, but has a sharp corner so the derivative is undefined).

8.2 Rules of Differentiation

Calculating limits manually is tedious. To differentiate efficiently, we use several established calculus rules.

Theorem/Rule: Calculus Differentiation Rules

Let f and g be differentiable functions, and let c be a constant.

- **Constant Rule:** $\frac{d}{dx}(c) = 0$
- **Power Rule:** $\frac{d}{dx}(x^n) = nx^{n-1} \quad \forall n \in \mathbb{R}$
- **Sum/Difference Rule:** $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$
- **Product Rule:** $\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x)$
- **Quotient Rule:** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
- **Chain Rule:** $\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$

Data Science Relevance (The Chain Rule): When you train a Deep Neural Network, the output is a composite function of many layers: $L(W_3(W_2(W_1(X))))$. To find out how a weight W_1 in the very first layer affects the final loss L , the algorithm uses the **Chain Rule** to multiply the derivatives backward through the network. This is called **Backpropagation**.

8.2.1 Derivatives of Standard Transcendental Functions

- **Exponential:** $\frac{d}{dx}(e^x) = e^x$
- **Logarithm:** $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$

Example: Differentiating with Chain and Quotient Rules

Find the derivative of $f(x) = \ln(x^2 + 3)$.

- Let the outer function be $u(y) = \ln(y)$ and the inner function be $y(x) = x^2 + 3$.
- Differentiate each component:

$$u'(y) = \frac{1}{y}, \quad y'(x) = 2x$$

- Apply the Chain Rule:

$$f'(x) = u'(y(x)) \cdot y'(x) = \frac{1}{x^2 + 3} \cdot (2x) = \frac{2x}{x^2 + 3}$$

Video Resource: 3Blue1Brown: Derivative Rules Geometrically



Presents brilliant visual explanations of the product, quotient, and chain rules using sliding area segments.

8.3 L'Hôpital's Rule

L'Hôpital's Rule provides an elegant method for evaluating limits that result in indeterminate forms (like $\frac{0}{0}$ or $\frac{\infty}{\infty}$).

Theorem/Rule: L'Hôpital's Rule

If $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$ (or if both limits are $\pm\infty$), then:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists (or is $\pm\infty$).

Example: Evaluating Indeterminate Limits

Find the limit: $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x}$.

- Direct substitution yields $\frac{e^0 - 1}{0} = \frac{1 - 1}{0} = \frac{0}{0}$, which is an indeterminate form.
- Apply L'Hôpital's Rule by differentiating the numerator and the denominator separately:

$$\text{Numerator derivative: } \frac{d}{dx}(e^{3x} - 1) = 3e^{3x}$$

$$\text{Denominator derivative: } \frac{d}{dx}(x) = 1$$

- Find the limit of the new quotient:

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x} = \lim_{x \rightarrow 0} \frac{3e^{3x}}{1} = 3e^0 = 3$$

8.4 Applications of Derivatives

8.4.1 Optimization: Maxima and Minima

In data science, we always want to find the minimum of a cost function (to reduce errors) or the maximum of a likelihood function (to increase accuracy). Derivatives allow us to find these extrema analytically.

Definition: Critical Points and Extrema Tests

- **Critical Point:** A value c in the domain of f where $f'(c) = 0$ (horizontal tangent) or $f'(c)$ is undefined.
- **First Derivative Test:** Let c be a critical point of a continuous function f :
 - If $f'(x)$ changes from positive (sloping up) to negative (sloping down) at c , then f has a **local maximum** at c .
 - If $f'(x)$ changes from negative to positive at c , then f has a **local minimum** at c .
- **Second Derivative Test:** Suppose $f'(c) = 0$ and $f''(c)$ exists:
 - If $f''(c) < 0$ (concave down), then f has a **local maximum** at c .
 - If $f''(c) > 0$ (concave up), then f has a **local minimum** at c .
 - If $f''(c) = 0$, the test is inconclusive (use the First Derivative Test).

Data Science Relevance (Gradient Descent): Instead of solving $f'(x) = 0$ analytically (which is impossible for neural networks with billions of parameters), algorithms use **Gradient Descent**. They compute the derivative at the current point, and take a small step in the *opposite* direction of the slope, physically walking down the curve until they hit a minimum (where $f'(x) = 0$).

```

1 def f_prime(x):
2     # Derivative of a simple loss function f(x) = x^2
3     return 2 * x
4
5 x_current = 5.0 # Starting guess
6 learning_rate = 0.1
7
8 # Take one step down the gradient
9 step = learning_rate * f_prime(x_current)
10 x_new = x_current - step
11
12 print(f"Old x: {x_current}, New x: {x_new}")
13 # The value moved closer to 0 (the minimum of x^2)

```

Listing 8.1: Data Science: Gradient Descent Step in Python

Example: Optimization Problem

Find the dimensions of a rectangle with a fixed perimeter of 100 meters that produces the maximum area.

1. Let the width be x and the length be y . The perimeter is:

$$2x + 2y = 100 \implies y = 50 - x$$

2. The area A is:

$$A(x) = x \cdot y = x(50 - x) = 50x - x^2$$

We restrict $x \in (0, 50)$ since physical dimensions must be positive.

3. Find the critical points by setting the derivative to zero:

$$A'(x) = 50 - 2x = 0 \implies 2x = 50 \implies x = 25$$

4. Apply the Second Derivative Test to confirm it's a maximum:

$$A''(x) = -2$$

Since $A''(25) = -2 < 0$, the function achieves a local (and global) **maximum** at $x = 25$.

5. Find the length y :

$$y = 50 - 25 = 25$$

So, the rectangle maximizing area is a **square** of dimensions $25\text{m} \times 25\text{m}$, yielding an area of 625 m^2 .

8.5 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Differentiate $f(x) = x^3e^x$ using the Product Rule.
2. **Data Science:** A machine learning Mean Squared Error cost function for a single parameter w is given by $C(w) = (w - 3)^2 + 4$. Find the critical point w where the gradient (derivative) is zero. Is this a maximum or minimum?
3. **Pure Math:** Find the derivative of $g(x) = \ln(3x^4 - 2x)$ using the Chain Rule.
4. **Pure Math:** Use L'Hôpital's Rule to evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.
5. **Data Science:** A simple neural network maps input x through two layers: $y = (2x + 1)^3$. Using the Chain Rule, what is $\frac{dy}{dx}$?

Chapter 9

Integration

Integration is the inverse operation of differentiation. While differentiation measures the instantaneous slope of a curve, integration calculates the continuous accumulation of quantities, geometrically represented as the area under a curve. In data science, integration is the mathematical cornerstone for probability. Specifically, it is used to calculate cumulative probability distributions, find expected values (means) of continuous random variables, and determine expected loss over distributions.

9.1 Riemann Sums and the Definite Integral

Consider a continuous function $f(x)$ defined on a closed interval $[a, b]$. We want to find the exact area bounded by the curve $y = f(x)$, the x-axis, and the vertical boundaries $x = a$ and $x = b$.

Definition: Riemann Sums and Definite Integral

- **Partition:** Divide the interval $[a, b]$ into n subintervals of equal width:

$$\Delta x = \frac{b - a}{n}$$

- **Riemann Sum:** Approximate the area using n rectangles. The sum of the areas of these rectangles is:

$$\text{Riemann Sum} = \sum_{i=1}^n f(x_i^*) \Delta x$$

where x_i^* is a sample point in the i -th subinterval $[x_{i-1}, x_i]$.

- **Definite Integral:** The definite integral of f from a to b is defined as the limit of the Riemann sum as the number of rectangles approaches infinity ($n \rightarrow \infty$):

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

The numbers a and b are called the lower and upper limits of integration.

Data Science Relevance: In statistics, a Probability Density Function (PDF), $p(x)$, describes the likelihood of a continuous random variable taking on a certain value. The probability that the variable falls between a and b is exactly the area under the curve between those points, calculated as $\int_a^b p(x) dx$.

9.2 The Fundamental Theorem of Calculus (FTC)

The FTC is one of the most remarkable discoveries in mathematics because it connects the two seemingly unrelated concepts of differentiation (slopes) and integration (areas).

Theorem/Rule: The Fundamental Theorem of Calculus

Let $f(x)$ be a continuous function on $[a, b]$.

- **FTC Part 1 (Derivative of an Integral):** If we define an area accumulation function $g(x) = \int_a^x f(t) dt$, then its rate of change is simply the curve's height:

$$g'(x) = \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$$

- **FTC Part 2 (Evaluation Theorem):** If $F(x)$ is any antiderivative of $f(x)$ (meaning $F'(x) = f(x)$), then the definite integral can be evaluated simply by subtracting the bounds:

$$\int_a^b f(x) dx = F(b) - F(a) = \left[F(x) \right]_a^b$$

9.3 Indefinite Integrals and Basic Techniques

An indefinite integral represents the family of all possible antiderivatives of a function. Because the derivative of a constant is zero, integration leaves ambiguity that we capture with $+C$.

$$\int f(x) dx = F(x) + C$$

Key Formula: Standard Integration Formulas

- **Power Rule:** $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (for $n \neq -1$)
- **Log Rule:** $\int \frac{1}{x} dx = \ln |x| + C$
- **Exponential Rule:** $\int e^x dx = e^x + C$
- **Constant Multiple Rule:** $\int c \cdot f(x) dx = c \int f(x) dx$
- **Sum Rule:** $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$

Example: Definite Integral Evaluation

Evaluate the definite integral: $\int_1^3 (3x^2 + 2x - 1) dx$.

1. Find the general antiderivative $F(x)$ using the Sum and Power Rules:

$$F(x) = \int (3x^2 + 2x - 1) dx = 3 \left(\frac{x^3}{3} \right) + 2 \left(\frac{x^2}{2} \right) - x = x^3 + x^2 - x$$

2. Evaluate $F(x)$ at the upper and lower limits:

$$F(3) = (3)^3 + (3)^2 - 3 = 27 + 9 - 3 = 33$$

$$F(1) = (1)^3 + (1)^2 - 1 = 1 + 1 - 1 = 1$$

3. Apply FTC Part 2 ($F(b) - F(a)$):

$$\int_1^3 (3x^2 + 2x - 1) dx = F(3) - F(1) = 33 - 1 = 32$$

9.3.1 The Substitution Rule (U-Substitution)

U-substitution is the "reverse Chain Rule". If a function and its derivative are both present inside the integral, we can compress it.

Theorem/Rule: The Substitution Rule

If $u = g(x)$ is a differentiable function, then $du = g'(x) dx$, allowing the transformation:

$$\int f(g(x))g'(x) dx = \int f(u) du$$

Example: Substitution Rule Application

Evaluate the indefinite integral: $\int 2xe^{x^2} dx$.

1. Let the inner exponent $u = x^2$. Differentiating with respect to x gives:

$$du = 2x dx$$

2. Substitute u and du entirely into the original integral:

$$\int e^{x^2} \cdot (2x dx) = \int e^u du$$

3. Integrate with respect to u :

$$\int e^u du = e^u + C$$

4. Substitute back $u = x^2$ to return to the original variable:

$$e^{x^2} + C$$

9.4 Area Between Curves

If $f(x) \geq g(x)$ for all x in $[a, b]$, the area A of the region bounded by the curves $y = f(x)$ on top, $y = g(x)$ on the bottom, and the vertical lines $x = a, x = b$ is simply the integral of the difference:

$$A = \int_a^b [f(x) - g(x)] dx$$

9.5 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Pure Math:** Evaluate the indefinite integral: $\int \left(4x^3 - \frac{1}{x} + e^x\right) dx$.
2. **Pure Math:** Use u-substitution to evaluate the integral: $\int x^2(x^3 + 5)^4 dx$.
3. **Data Science:** The Probability Density Function of a uniform distribution is given by $p(x) = 0.5$ on the interval $[0, 2]$. Verify that the total probability over this entire interval is exactly 1 by calculating $\int_0^2 0.5 dx$.
4. **Pure Math:** Evaluate the definite integral using FTC Part 2: $\int_1^2 \frac{1}{x^2} dx$.
5. **Pure Math:** Find the area of the region enclosed between the curves $y = x^2$ and $y = x$ on the interval $[0, 1]$.

Chapter 10

Graph Theory (I): BFS, DFS, and Topo Sort

Graph theory is the study of networks consisting of points (vertices) connected by lines (edges). In data science, graphs model relations between data points, such as social networks, recommendation systems, page ranking (PageRank), and dependencies between variables in probabilistic models (like Bayesian Networks).

10.1 Introduction and Classifications

A **graph** G is a pair $G = (V, E)$, where V is a non-empty set of vertices (or nodes) and E is a set of edges (connections between vertices).

Definition: Graph Classifications

- **Undirected Graph:** Edges have no orientation. An edge is an unordered pair $\{u, v\}$.
- **Directed Graph (Digraph):** Edges have orientation. An edge is an ordered pair (u, v) , representing a one-way flow from $u \rightarrow v$.
- **Simple Graph:** A graph with no loops (edges from a vertex to itself) and no multiple edges (parallel edges between the same two vertices).
- **Weighted Graph:** A graph where each edge is assigned a numerical weight/cost (like distance or time).

Video Resource: IITM BS Lecture: W10_L1 Introduction to Graphs 

Official IIT Madras lecture introducing vertices, edges, connectivity, and basic terminology.

10.2 Representations of Graphs

How we store graphs in memory dictates algorithm performance.

Definition: Graph Storage Representations

Let $G = (V, E)$ with $V = \{1, 2, \dots, n\}$.

- **Adjacency Matrix (A):** A square $n \times n$ matrix where $A_{ij} = 1$ if $(i, j) \in E$, and 0 otherwise.
 - For undirected graphs, A is symmetric ($A_{ij} = A_{ji}$).
 - Excellent for dense graphs, but wastes $O(|V|^2)$ memory on sparse graphs.
- **Adjacency List:** An array of lists. The list at index i contains all vertices adjacent to vertex i . This is highly space-efficient for **sparse graphs** (where $|E| \ll |V|^2$), requiring only $O(|V| + |E|)$ memory.

Video Resource: IITM BS Lecture: W10_L3 Representation of Graphs 

Official IIT Madras lecture comparing matrices and lists, detailing spatial complexity trade-offs.

10.3 Graph Traversals: BFS and DFS

To analyze or search a graph, we must visit all its vertices systematically.

Theorem/Rule: BFS and DFS Traversals

- **Breadth-First Search (BFS):**
 - **Concept:** Explores the graph level-by-level (radiating outward like ripples in a pond), visiting all neighbors of a node before moving to the next level.
 - **Data Structure:** Uses a **Queue** (First-In, First-Out).
 - **Application:** Finds the shortest path in an **unweighted** graph.
- **Depth-First Search (DFS):**
 - **Concept:** Explores as deep as possible along each branch before backtracking.
 - **Data Structure:** Uses a **Stack** (Last-In, First-Out) or **Recursion**.
 - **Application:** Cycle detection, finding connected components, maze solving.

Complexity: Both algorithms run in $O(|V| + |E|)$ time when using an Adjacency List.

Core Intuition: BFS vs DFS Traversals

- **BFS Intuition:** Imagine dropping a stone in a pond. The ripples explore everything 1 unit away, then 2 units away. It guarantees you find the shortest path.
- **DFS Intuition:** Imagine walking through a physical maze. You keep your hand on the right wall and walk as deep as you can until you hit a dead end, then backtrack.

Example: BFS Algorithm Trace

Consider a simple graph with vertices $\{0, 1, 2, 3\}$ and edges $\{(0, 1), (0, 2), (1, 2), (2, 3)\}$. Let's trace BFS starting at node 0.

Step	Current Node	Queue (Front to Back)	Visited Set
1	(Start)	[0]	{0}
2	Pop 0 Push neighbors 1, 2	[1, 2]	{0, 1, 2}
3	Pop 1 Neighbors already visited	[2]	{0, 1, 2}
4	Pop 2 Push neighbor 3	[3]	{0, 1, 2, 3}
5	Pop 3 Neighbors already visited	[]	{0, 1, 2, 3}

BFS completely explores level 1 (nodes 1, 2) before reaching level 2 (node 3).

Video Resource: IITM BS Lecture: W10_L4 Breadth-First Search (BFS)



Official IIT Madras lecture detailing the queue mechanism and level-order traversal of BFS.

Video Resource: IITM BS Lecture: W10_L5 Depth-First Search (DFS)



Official IIT Madras lecture detailing the recursive stack mechanism and backtracking in DFS.

10.4 Directed Acyclic Graphs (DAGs) and Topological Sorting

Definition: DAGs and Topological Sort

- **DAG:** A Directed Acyclic Graph is a directed graph that contains **no directed cycles**. If you start at a node and follow the arrows, you can never return to that node.
- **Topological Sorting:** A linear ordering of vertices such that for every directed edge $u \rightarrow v$, vertex u comes before v in the ordering. This is only possible if the graph is a DAG.

Data Science Relevance: In data engineering pipelines (like Apache Airflow), tasks have dependencies (e.g., Task B cannot start until Task A finishes). The pipeline is modeled as a DAG. Topological sorting determines the exact sequential order to execute the tasks so that no dependencies are broken.

Video Resource: IITM BS Lecture: W10_L9 Topological Sorting



Official IIT Madras lecture showing how to use DFS (or Kahn's Algorithm) to generate a valid topological order.

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W10_L2: Some general graph problems](#)
- [W10_L6: Applications of BFS & DFS | part 1](#)
- [W10_L7: Applications of BFS & DFS | part 2](#)
- [W10_L8: Complexity of BFS & DFS](#)

10.5 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Computer Science:** An undirected graph has 6 vertices and 15 edges. Is it a simple graph, or must it contain multiple edges/loops? (Hint: consider the maximum number of edges in a simple graph with n vertices).
2. **Data Structure:** A graph is represented by an Adjacency Matrix. What does it mean if the sum of all elements in the i -th row is 3? Assume the graph is unweighted and directed.
3. **Algorithms:** You are traversing a graph to find the shortest path out of a maze. The graph is unweighted. Should you use BFS or DFS, and why?
4. **Algorithms:** You have a DAG representing university course prerequisites. You want to generate a valid semester-by-semester graduation plan. Which graph algorithm should you use?
5. **Complexity:** An undirected graph $G = (V, E)$ has $|V| = 10,000$ and $|E| = 20,000$. Which storage representation is more memory efficient: an Adjacency Matrix or an Adjacency List? Justify mathematically.

Chapter 11

Graph Theory (II): Shortest Paths and Spanning Trees

This chapter explores advanced structural graph algorithms, focusing on heavily weighted networks. We will learn how to find the most efficient paths between nodes and how to connect an entire network with the minimum possible cost. In data science and operations research, these algorithms solve massive logistics problems, network routing (like Google Maps), and cluster optimization.

11.1 Transitive Closure and Reachability

The **transitive closure** of a directed graph $G = (V, E)$ determines whether there exists a directed path from vertex i to vertex j for all pairs $i, j \in V$.

Theorem/Rule: Warshall's Algorithm

Warshall's algorithm is a dynamic programming approach that constructs a boolean reachability matrix $R^{(k)}$ where $R_{ij}^{(k)} = 1$ if there is a path from $i \rightarrow j$ using only intermediate vertices from the set $\{1, 2, \dots, k\}$.

- **Recurrence Relation:**

$$R_{ij}^{(k)} = R_{ij}^{(k-1)} \vee \left(R_{ik}^{(k-1)} \wedge R_{kj}^{(k-1)} \right)$$

- **Time Complexity:** $\mathcal{O}(|V|^3)$

Data Science Relevance: In social networks, if User A follows User B, and User B follows User C, the transitive closure tells the recommendation engine that User A might be interested in User C (a path exists).

Video Resource: IITM BS Lecture: W11_L2 Transitive Closure & Reachability



Official IIT Madras lecture detailing the structural concept of reachability matrices.

11.2 Shortest Path Algorithms

In weighted graphs, BFS no longer finds the shortest path because a path with more edges might have a lower total weight. We need specialized algorithms.

11.2.1 Single-Source Shortest Paths

These algorithms find the shortest path from a *single starting vertex* to all other vertices in the graph.

Definition: Dijkstra's Algorithm

Dijkstra's Algorithm uses a greedy approach to find the shortest paths from a source node S .

- **Mechanism:** Maintains a set of visited nodes and a priority queue of unvisited nodes, always expanding the node with the current shortest known distance from S .
- **Constraint:** Fails if the graph contains **negative edge weights**.
- **Complexity:** $\mathcal{O}(|E| \log |V|)$ using a min-heap priority queue.

Example: Dijkstra's Algorithm Conceptual Steps

Suppose we want to find the shortest path from start node S to all other nodes.

1. **Initialization:** Set the distance to S to 0. Set the distance to all other nodes to ∞ . Mark all nodes as unvisited.
2. **Selection:** Pick the unvisited node with the smallest known distance. Initially, this is S (distance 0).
3. **Relaxation:** For the current node, examine all its unvisited neighbors. Calculate the tentative distance to each neighbor through the current node. If this tentative distance is smaller than the neighbor's currently known distance, update the neighbor's distance.
4. **Completion:** Mark the current node as visited. A visited node's shortest distance is now finalized and will never change (assuming no negative edges).
5. **Iteration:** Repeat steps 2-4 until all nodes are visited, or until the smallest known distance among unvisited nodes is ∞ (meaning the remaining nodes are unreachable).

Video Resource: IITM BS Lecture: W11_L5 Dijkstra's Algorithm



Official IIT Madras lecture tracing Dijkstra's greedy path selection step-by-step.

Definition: Bellman-Ford Algorithm

Bellman-Ford also finds single-source shortest paths, but is more robust than Dijkstra's.

- **Mechanism:** Relaxes (updates) *all* edges $|V| - 1$ times. If distances still update on the $|V|$ -th pass, it proves the graph has a negative weight cycle.
- **Advantage:** Works perfectly with **negative edge weights** and detects negative cycles.
- **Complexity:** $\mathcal{O}(|V| \cdot |E|)$ (slower than Dijkstra).

Video Resource: IITM BS Lecture: W11_L6 Bellman-Ford Algorithm



Official IIT Madras lecture explaining edge relaxation and negative cycle detection.

11.2.2 All-Pairs Shortest Paths

If we want the shortest path between *every* pair of vertices, we use the **Floyd-Warshall Algorithm**. It is a dynamic programming algorithm (an extension of Warshall's) that evaluates intermediate nodes to find all shortest paths in $\mathcal{O}(|V|^3)$ time.

11.3 Minimum Cost Spanning Trees (MST)

Definition: Spanning Tree

Given a connected, undirected graph $G = (V, E)$, a **spanning tree** is a subgraph that is a tree (contains no cycles) and includes every single vertex from V . It will always have exactly $|V| - 1$ edges.

If the graph is weighted, the **Minimum Spanning Tree (MST)** is the spanning tree whose total edge weight is as small as possible. This is used in network design (e.g., laying the minimum length of fiber optic cable to connect a series of cities).

Theorem/Rule: Algorithms for Finding the MST

- **Prim's Algorithm (Node-based):**
 - Start at an arbitrary node.
 - Greedily add the cheapest edge that connects a visited node to an unvisited node.
 - Repeat until all nodes are visited.
- **Kruskal's Algorithm (Edge-based):**
 - Sort all edges in the graph from lightest to heaviest.
 - Iterate through the sorted edges, adding them to the MST *only if* they do not create a cycle (usually tracked via a Disjoint-Set / Union-Find data structure).

Video Resource: IITM BS Lecture: W11_L8 Minimum Cost Spanning Trees 

Official IIT Madras lecture introducing the MST problem and comparing the mechanics of Prim and Kruskal.

Example: NetworkX Implementation: Dijkstra & Kruskal MST

In Data Science, graph algorithms are efficiently executed using the `networkx` Python library:

```
1 import networkx as nx
2
3 # Build a weighted graph
4 G = nx.Graph()
5 edges = [
6     ('A', 'B', 4), ('A', 'C', 2),
7     ('B', 'C', 1), ('B', 'D', 5),
8     ('C', 'D', 8), ('C', 'E', 10),
9     ('D', 'E', 2)
10 ]
11 for u, v, w in edges:
12     G.add_edge(u, v, weight=w)
13
14 # 1. Single-Source Shortest Path (Dijkstra)
15 path = nx.dijkstra_path(G, source='A', target='E')
16 length = nx.dijkstra_path_length(G, source='A', target='E')
17 print(f"Shortest path A -> E: {path} with weight {length}")
18
19 # 2. Minimum Spanning Tree (Kruskal)
20 mst = nx.minimum_spanning_tree(G, algorithm='kruskal')
21 print("MST Edges:", list(mst.edges(data=True)))
```

Supplementary Lecture Videos

For further reinforcement and specialized topics, refer to the following official IIT Madras lectures:

- [W11_L1: Longest paths in DAGs task scheduling & dependencies](#)
- [W11_L3: Matrix multiplication | graphs, boolean algebra & transitive closure](#)
- [W11_L4: Shortest paths in weighted graphs | single source & all pairs](#)
- [W11_L7: All-pairs shortest paths | floyd-warshall algorithm](#)
- [W11_L9: Minimum cost spanning trees | prims algorithm](#)
- [W11_L10: Minimum cost spanning trees | kruskals algorithm](#)

11.4 Practice Exercises

Try to solve these exercises independently. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Algorithms:** You are writing navigation software for a delivery drone. The drone battery drains faster if it flies into a headwind, which is modeled as an edge with a negative weight (cost). Which shortest-path algorithm *must* you use?
2. **Algorithms:** An undirected graph has 8 vertices. How many edges will its Minimum Spanning Tree have?
3. **Pure Math:** You apply Kruskal's algorithm to a graph. The next lightest edge connects Node A and Node B. However, Node A and Node B are already in the same connected component of your growing MST. Should you add the edge? Why or why not?
4. **Complexity:** Why is Floyd-Warshall ($\mathcal{O}(|V|^3)$) preferred over running Dijkstra's algorithm from every single node ($\mathcal{O}(|V| \cdot |E| \log |V|)$) for dense graphs when finding all-pairs shortest paths?
5. **Graph Theory:** What mathematical operation on an Adjacency Matrix A calculates the number of walks of exactly length k between any two nodes?

Chapter 12

Course Revision and Final Review

Congratulations on reaching the final week! This chapter serves as a comprehensive synthesis of the entire Maths 1 curriculum. We will review the core conceptual pillars that form the foundation for all subsequent Data Science and Machine Learning coursework.

12.1 Core Concept Cheat Sheet

12.1.1 1. Set Theory and Relations

- **Sets:** Collections of distinct objects. Operations include Union ($\cup$), Intersection ($\cap$), Difference ($-$), and Complement (A^c).
- **Relations:** Subsets of Cartesian products ($A \times B$). Equivalence relations must be *Reflexive*, *Symmetric*, and *Transitive*.
- **Data Science Link:** Relational databases (SQL) are entirely built on set theory operations (e.g., INNER JOIN is an intersection).

12.1.2 2. Functions and Polynomials

- **Functions:** A mapping where each input has exactly one output. Checked via the Vertical Line Test.
- **Polynomials:** $P(x) = a_n x^n + \dots + a_0$. The degree determines the maximum number of roots (n) and turning points ($n - 1$).
- **Data Science Link:** Polynomial Regression captures non-linear trends, but high-degree polynomials risk "overfitting" the training data.

12.1.3 3. Exponentials and Logarithms

- **Exponentials (e^x):** Represent rapid, compounding growth.
- **Logarithms ($\ln x$):** The inverse of exponentials. They scale massive numbers down and turn multiplication into addition ($\ln(xy) = \ln(x) + \ln(y)$).
- **Data Science Link:** Log transformations normalize skewed datasets. The natural logarithm is the engine of the Cross-Entropy Loss function in classification tasks.

12.1.4 4. Calculus (Limits and Derivatives)

- **Derivative ($f'(x)$):** The instantaneous rate of change (slope of the tangent line), defined by the limit $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$.
- **Chain Rule:** $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$.
- **Data Science Link:** The Chain Rule is the exact mechanism of **Backpropagation**, the algorithm used to train deep neural networks via **Gradient Descent** (finding where $f'(x) = 0$).

12.1.5 5. Graph Theory and Algorithms

- **Representations:** Adjacency Matrices (dense) vs. Adjacency Lists (sparse).
- **Algorithms:** BFS (shortest paths unweighted), DFS (cycles/topological sort), Dijkstra (shortest paths positive weights), Kruskal/Prim (Minimum Spanning Trees).
- **Data Science Link:** Directed Acyclic Graphs (DAGs) model workflow dependencies (Apache Airflow) and causal networks.

12.2 Official IITM Revision Lectures

To solidify your understanding before the final exam, the official IIT Madras curriculum provides dedicated revision lectures for every core topic.

Video Resource: IITM BS Lecture: W12_L4 Revision on Functions



A complete review of domains, ranges, injectivity, and mapping types.

Video Resource: IITM BS Lecture: W12_L3 Revision on Polynomials



A complete review of roots, turning points, and end behavior visualization.

Video Resource: IITM BS Lecture: W12_L5 Revision on Logarithms



A complete review of logarithmic laws, equation solving, and inverse properties.

Video Resource: IITM BS Lecture: W12_L6 Revision on Graph Theory



A complete review of adjacency matrices, topological sorting, and shortest path logic.

Supplementary Revision Lectures

- [W12_L1: Revision lecture on set theory | venn diagrams, operations](#)
- [W12_L2: Revision lecture on relations | properties, representations](#)
- [W12_L7: Revision lecture on exponential & composite function](#)

12.3 Comprehensive Final Practice Exam

Test your mastery of the entire course. Detailed step-by-step solutions are available in the Appendix at the end of the book.

1. **Sets & Logic:** In a dataset of 100 users, 60 like Apples (A), 50 like Bananas (B), and 20 like neither. How many users like *both* Apples and Bananas?
2. **Functions & Exponentials:** Given $f(x) = e^{2x}$ and $g(x) = \ln(x)$, find the composite function $(f \circ g)(x)$ and simplify it completely.
3. **Calculus:** Find the critical points of the loss function $L(w) = w^3 - 3w^2 - 9w + 15$. Use the Second Derivative Test to determine which point is a local minimum (the optimal weight).
4. **Integration:** Calculate the total probability area under the curve $p(x) = 3x^2$ between $x = 0$ and $x = 1$.
5. **Graph Theory:** You are using Dijkstra's algorithm to find the fastest driving route. You discover a toll road that pays you \$5 to drive on it (a negative weight of -5). Will Dijkstra's algorithm still guarantee the correct shortest path? What algorithm should you use instead?

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Solutions to Practice Exercises

This appendix provides detailed, step-by-step solutions for the practice exercises found at the end of each chapter. The exercises are designed to blend pure mathematical reasoning with practical data science applications.

Week 1: Sets, Relations, and Functions

Detailed Solutions

1. Pure Math: Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Find $(A \cup B) \setminus (A \cap B)$.

- First, find the union $A \cup B$, which represents all elements in either set: $A \cup B = \{1, 2, 3, 4, 5\}$.
- Next, find the intersection $A \cap B$, which represents elements present in both sets: $A \cap B = \{3\}$.
- Finally, take the difference, which removes elements of the intersection from the union: $(A \cup B) \setminus (A \cap B) = \{1, 2, 4, 5\}$.

2. Data Science: Express "Users who clicked the ad but did not buy the product" using set notation.

- Let C be the set of users who clicked the ad.
- Let B be the set of users who bought the product.
- We want elements in C that are NOT in B . This is the definition of set difference.
- Therefore, the expression is $C \setminus B$ (or $C - B$).

3. Pure Math: Determine if the relation R on $\mathbb{R}$ defined by $xRy \iff x < y$ is an equivalence relation.

- **Reflexivity:** Is $x < x$ for all real numbers x ? No. For example, $5 \not< 5$. Since reflexivity fails, R cannot be an equivalence relation.
- **Symmetry:** If $x < y$, is $y < x$? No. If $3 < 5$, $5 \not< 3$. Symmetry fails.
- **Transitivity:** If $x < y$ and $y < z$, then $x < z$. Yes, this property holds.
- **Conclusion:** It is not an equivalence relation because it is neither reflexive nor symmetric.

4. Data Science: Is the relation "lives in the same city as" an equivalence relation?

- **Reflexivity:** Does User A live in the same city as User A? Yes.
- **Symmetry:** If User A lives in the same city as User B, does User B live in the same city as User A? Yes.

- **Transitivity:** If User A lives in the same city as User B, and User B lives in the same city as User C, does User A live in the same city as User C? Yes.
- **Conclusion:** Since it satisfies all three properties, it is an equivalence relation. It perfectly partitions users into disjoint sets (cities).

5. Pure Math: Prove whether the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2$ is injective, surjective, both, or neither.

- **Injectivity:** Does $f(x_1) = f(x_2)$ imply $x_1 = x_2$? Let's test negative numbers. $f(2) = 4$ and $f(-2) = 4$. Since $2 \neq -2$ but their outputs are the same, the function is NOT injective.
- **Surjectivity:** For every integer y , is there an integer x such that $x^2 = y$? Consider $y = -1$ (which is in the codomain $\mathbb{Z}$). There is no integer x such that $x^2 = -1$. The function is NOT surjective.
- **Conclusion:** The function is neither injective nor surjective.

6. Data Science: Prove that the Min-Max scaling function $g: [0, 10] \rightarrow [0, 1]$ defined by $g(x) = \frac{x}{10}$ is a bijection.

- **Injectivity:** Assume $g(x_1) = g(x_2)$.

$$\frac{x_1}{10} = \frac{x_2}{10} \implies x_1 = x_2$$

Thus, g is injective.

- **Surjectivity:** Let $y \in [0, 1]$. We need to find an $x \in [0, 10]$ such that $g(x) = y$.

$$\frac{x}{10} = y \implies x = 10y$$

Since $0 \leq y \leq 1$, it follows that $0 \leq 10y \leq 10$. Therefore, $x = 10y$ is a valid input in the domain $[0, 10]$. Check: $g(10y) = \frac{10y}{10} = y$. Thus, g is surjective.

- **Conclusion:** Since it is both injective and surjective, g is a bijection. This guarantees that our data scaling is a perfectly reversible operation.

Week 2: Coordinate Geometry and Straight Lines

Detailed Solutions

1. Pure Math: Find the distance between the points $A(3, 4)$ and $B(7, 1)$.

- Use the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- $d = \sqrt{(7 - 3)^2 + (1 - 4)^2} = \sqrt{4^2 + (-3)^2}$.
- $d = \sqrt{16 + 9} = \sqrt{25} = 5$.

2. Pure Math: A point P divides the line segment joining $X(1, -2)$ and $Y(-3, 6)$ internally in the ratio $1 : 3$. Find the coordinates of P .

- Use the internal section formula with $m = 1, n = 3, (x_1, y_1) = (1, -2)$, and $(x_2, y_2) = (-3, 6)$.
- $x = \frac{1(-3) + 3(1)}{1+3} = \frac{-3+3}{4} = 0$.
- $y = \frac{1(6) + 3(-2)}{1+3} = \frac{6-6}{4} = 0$.

- The coordinates of point P are $(0, 0)$.

3. Data Science: A K-Means clustering algorithm places a centroid at $(5, 5)$. A new data point arrives at $(2, 9)$. What is the Euclidean distance from the point to the centroid?

- Euclidean distance is exactly the distance formula.

- $d = \sqrt{(2-5)^2 + (9-5)^2} = \sqrt{(-3)^2 + 4^2}$.

- $d = \sqrt{9 + 16} = \sqrt{25} = 5$.

4. Pure Math: Find the equation of the line passing through the points $(2, 3)$ and $(4, 7)$ in slope-intercept form.

- First find the slope $m = \frac{7-3}{4-2} = \frac{4}{2} = 2$.

- Use point-slope form: $y - y_1 = m(x - x_1) \implies y - 3 = 2(x - 2)$.

- Simplify to slope-intercept form ($y = mx + c$): $y = 2x - 4 + 3 \implies y = 2x - 1$.

5. Data Science: A simple linear regression model predicts housing prices (y) based on square footage (x). The model parameters are learned as slope $m = 150$ and intercept $c = 50000$. If a house has 2000 square feet, what is the predicted price?

- The model equation is $y = mx + c = 150x + 50000$.

- Substitute $x = 2000$: $y = 150(2000) + 50000 = 300000 + 50000$.

- The predicted price is 350,000.

6. Pure Math: Find the perpendicular distance from the origin $(0, 0)$ to the line $3x - 4y + 15 = 0$.

- Use the point-to-line distance formula: $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.

- Here $(x_0, y_0) = (0, 0)$ and the line is $3x - 4y + 15 = 0$ ($A = 3, B = -4, C = 15$).

- $d = \frac{|3(0) - 4(0) + 15|}{\sqrt{3^2 + (-4)^2}} = \frac{15}{\sqrt{9+16}} = \frac{15}{5} = 3$.

Week 3: Quadratic Functions

Detailed Solutions

1. Pure Math: Find the roots of the quadratic equation $3x^2 - 5x - 2 = 0$ using factorization.

- We need two numbers that multiply to $(3)(-2) = -6$ and add to -5 . These numbers are -6 and 1 .

- Rewrite the middle term: $3x^2 - 6x + x - 2 = 0$.

- Factor by grouping: $3x(x - 2) + 1(x - 2) = 0 \implies (3x + 1)(x - 2) = 0$.

- The roots are $x = -\frac{1}{3}$ and $x = 2$.

2. Pure Math: Calculate the discriminant of $4x^2 + 12x + 9 = 0$ and state the nature of its roots.

- The discriminant is $D = b^2 - 4ac$. Here, $a = 4, b = 12, c = 9$.

- $D = (12)^2 - 4(4)(9) = 144 - 144 = 0$.
- Since $D = 0$, the equation has one repeated real root.

3. Data Science: A Mean Squared Error loss function for a single parameter w is defined as $L(w) = 2w^2 - 16w + 35$. Find the value of w that minimizes the loss, and the minimum loss value itself.

- This is an optimization problem where $a = 2, b = -16, c = 35$. Since $a > 0$, it is a parabola opening upward (a minimum exists).
- The minimum occurs at the vertex: $w = -\frac{b}{2a} = -\frac{-16}{2(2)} = \frac{16}{4} = 4$.
- The minimum loss value is $L(4) = 2(4)^2 - 16(4) + 35 = 32 - 64 + 35 = 3$.
- So, setting parameter $w = 4$ minimizes the loss to 3.

4. Pure Math: Convert the quadratic function $y = x^2 + 6x + 5$ into vertex form and identify its vertex.

- The vertex form is $y = a(x - h)^2 + k$. We complete the square.
- Half of the x coefficient is $\frac{6}{2} = 3$, and $3^2 = 9$.
- Add and subtract 9: $y = (x^2 + 6x + 9) - 9 + 5$.
- $y = (x + 3)^2 - 4$.
- The vertex (h, k) is $(-3, -4)$.

5. Data Science: A manufacturing algorithm predicts that producing n items yields a cost $C(n) = 0.5n^2 - 50n + 1500$. How many items should be produced to minimize the cost?

- Here $a = 0.5, b = -50$. Since $a > 0$, this parabola has a minimum.
- The number of items that minimizes cost is given by the vertex x-coordinate: $n = -\frac{b}{2a}$.
- $n = -\frac{-50}{2(0.5)} = \frac{50}{1} = 50$.
- Producing 50 items minimizes the cost.

Week 4: Algebra of Polynomials

Detailed Solutions

1. Pure Math: Use the Remainder Theorem to find the remainder when $P(x) = x^4 - 2x^3 + 5x - 7$ is divided by $(x - 1)$.

- By the Remainder Theorem, the remainder is $P(1)$.
- Substitute $x = 1$: $P(1) = (1)^4 - 2(1)^3 + 5(1) - 7$.
- $P(1) = 1 - 2 + 5 - 7 = 6 - 9 = -3$.
- The remainder is -3 .

2. Pure Math: Determine the end behavior of the polynomial $f(x) = 3x^5 - 4x^3 + 2x - 1$.

- The leading term is $3x^5$.

- The degree $n = 5$ is odd, and the leading coefficient $a_n = 3$ is positive.
- According to the Leading Coefficient Test for odd degree/positive coefficient, the end behavior is Down-Up.
- As $x \rightarrow \infty, y \rightarrow \infty$, and as $x \rightarrow -\infty, y \rightarrow -\infty$.

3. Data Science: You are fitting a polynomial regression curve to 4 data points. What is the maximum number of turning points a degree-3 polynomial curve can have as it passes through the data?

- A polynomial of degree n has at most $n - 1$ turning points.
- For $n = 3$, the maximum number of turning points is $3 - 1 = 2$.

4. Pure Math: Find the x-intercepts and state their multiplicities for the polynomial $g(x) = x^2(x + 3)^3(x - 5)$. Will the graph cross or bounce at $x = -3$?

- Set $g(x) = 0$ to find the roots:
- $x = 0$ has multiplicity 2 (even).
- $x = -3$ has multiplicity 3 (odd).
- $x = 5$ has multiplicity 1 (odd).
- At $x = -3$, because the multiplicity is odd, the graph will **cross** the x-axis (with flattening, since $3 > 1$).

5. Data Science: A complex machine learning boundary is defined by the polynomial equation $B(x) = -2x^6 + 5x^4 - 3x^2 + 1$. As $x \rightarrow \infty$ and $x \rightarrow -\infty$, in which direction does the boundary eventually head?

- Look at the leading term: $-2x^6$.
- The degree $n = 6$ is even, and the leading coefficient $a_n = -2$ is negative.
- By the Leading Coefficient Test for even degree/negative coefficient, the end behavior is Down-Down.
- As $x \rightarrow \infty, B(x) \rightarrow -\infty$, and as $x \rightarrow -\infty, B(x) \rightarrow -\infty$.

Week 5: Functions, Compositions, and Exponentials

Detailed Solutions

1. Pure Math: Use the horizontal line test conceptually to determine if $f(x) = x^2$ is one-to-one on the domain $\mathbb{R}$. What if the domain is restricted to $[0, \infty)$?

- On the domain $\mathbb{R}$, the graph of $f(x) = x^2$ is a full parabola opening upwards. A horizontal line like $y = 4$ intersects the graph at two points: $(-2, 4)$ and $(2, 4)$. Thus, it is **not** one-to-one on $\mathbb{R}$.
- If restricted to $[0, \infty)$, we only have the right half of the parabola. Any horizontal line $y > 0$ will intersect the graph exactly once. Thus, it is one-to-one on $[0, \infty)$.

2. Pure Math: Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Are they equal?

- $(f \circ g)(x) = f(g(x)) = f(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10$.
- $(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = 2(x^2 + 1) - 3 = 2x^2 + 2 - 3 = 2x^2 - 1$.
- $4x^2 - 12x + 10 \neq 2x^2 - 1$. Thus, function composition is **not** commutative in general.

3. Data Science: In a neural network, a linear transformation $L(x) = 3x - 2$ is followed by a ReLU activation function $R(x) = \max(0, x)$. Evaluate the composite output $(R \circ L)(1)$ and $(R \circ L)(0)$.

- For $x = 1$: First, $L(1) = 3(1) - 2 = 1$. Then, $R(1) = \max(0, 1) = 1$. So, $(R \circ L)(1) = 1$.
- For $x = 0$: First, $L(0) = 3(0) - 2 = -2$. Then, $R(-2) = \max(0, -2) = 0$. So, $(R \circ L)(0) = 0$.
- This composite structure $(R(L(x)))$ acts as an "activation" switch, only passing positive signals forward.

4. Pure Math: Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{4x}{x-2}$.

- Set $y = \frac{4x}{x-2}$.
- Multiply by $(x - 2)$: $y(x - 2) = 4x \implies xy - 2y = 4x$.
- Collect x terms on one side: $xy - 4x = 2y \implies x(y - 4) = 2y$.
- Solve for x : $x = \frac{2y}{y-4}$.
- Swap variables: $f^{-1}(x) = \frac{2x}{x-4}$.

5. Pure Math: Simplify the exponential expression: $\frac{(e^{2x})^3 \cdot e^{-x}}{e^{4x}}$.

- Apply power of a power rule: $(e^{2x})^3 = e^{6x}$.
- The expression becomes: $\frac{e^{6x} \cdot e^{-x}}{e^{4x}}$.
- Apply multiplication rule (add exponents) in numerator: $e^{6x+(-x)} = e^{5x}$.
- Apply division rule (subtract exponents): $\frac{e^{5x}}{e^{4x}} = e^{5x-4x} = e^x$.
- The simplified expression is simply e^x .

Week 6: Logarithmic Functions

Detailed Solutions

1. Pure Math: Evaluate the expression without a calculator: $\log_3(81) - \log_2\left(\frac{1}{8}\right)$.

- Evaluate the first term: $3^4 = 81$, so $\log_3(81) = 4$.
- Evaluate the second term: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$, so $\log_2\left(\frac{1}{8}\right) = -3$.
- Subtract the values: $4 - (-3) = 4 + 3 = 7$.

2. Data Science: In binary classification, the Cross-Entropy loss for a true label $y = 1$ and predicted probability p is defined as $L = -\ln(p)$. If the predicted probability is $p = e^{-2}$, what is the loss?

- Substitute p into the loss function: $L = -\ln(e^{-2})$.

- Use the inverse property $\ln(e^x) = x$: $L = -(-2) = 2$.
- The loss is 2.

3. Pure Math: Use the laws of logarithms to expand the expression completely: $\ln\left(\frac{x^4\sqrt{y^3}}{e^2}\right)$.

- First, apply the quotient rule: $\ln(x^4\sqrt{y^3}) - \ln(e^2)$.
- Apply the product rule to the first term: $\ln(x^4) + \ln(y^{3/2}) - \ln(e^2)$.
- Apply the power rule to all terms, and use $\ln(e) = 1$: $4\ln(x) + \frac{3}{2}\ln(y) - 2(1)$.
- The expanded form is: $4\ln(x) + \frac{3}{2}\ln(y) - 2$.

4. Data Science: A dataset feature X ranges from 1 to 1,000,000. A data scientist applies a base-10 log transformation to create a new feature $X' = \log_{10}(X)$. What is the new range of X' ?

- Find the minimum value of X' : $\log_{10}(1) = 0$.
- Find the maximum value of X' : $\log_{10}(1,000,000) = \log_{10}(10^6) = 6$.
- The new range of X' is incredibly compressed, spanning from 0 to 6.

5. Pure Math: Solve the equation for x : $e^{2x} - 3e^x + 2 = 0$. (Hint: treat it as a quadratic in form).

- Let $u = e^x$. The equation becomes a quadratic: $u^2 - 3u + 2 = 0$.
- Factor the quadratic: $(u - 1)(u - 2) = 0$.
- So, $u = 1$ or $u = 2$.
- Substitute back $e^x = u$: $e^x = 1 \implies x = \ln(1) = 0$.
- And $e^x = 2 \implies x = \ln(2)$.
- The solutions are $x = 0$ and $x = \ln(2)$.

6. Pure Math: Solve the logarithmic equation: $\log(x + 3) - \log(x) = 1$.

- (Note: $\log$ without a base usually means base 10). Apply the quotient rule: $\log\left(\frac{x+3}{x}\right) = 1$.
- Convert to exponential form: $\frac{x+3}{x} = 10^1 = 10$.
- Multiply by x : $x + 3 = 10x$.
- Solve for x : $3 = 9x \implies x = \frac{3}{9} = \frac{1}{3}$.
- Check validity: Both $x + 3 = \frac{10}{3} > 0$ and $x = \frac{1}{3} > 0$ are valid inside the logs. The solution is $x = \frac{1}{3}$.

Week 7: Sequences and Limits (Part 1)

Detailed Solutions

1. Pure Math: Determine if the sequence $b_n = \frac{5n^3 - n + 2}{7n^3 + 4n^2}$ converges, and if so, find its limit.

- Divide numerator and denominator by the highest power of n , which is n^3 :

$$b_n = \frac{\frac{5n^3}{n^3} - \frac{n}{n^3} + \frac{2}{n^3}}{\frac{7n^3}{n^3} + \frac{4n^2}{n^3}} = \frac{5 - \frac{1}{n^2} + \frac{2}{n^3}}{7 + \frac{4}{n}}$$

- Take the limit as $n \rightarrow \infty$. All fraction terms with n in the denominator go to 0:

$$\lim_{n \rightarrow \infty} b_n = \frac{5 - 0 + 0}{7 + 0} = \frac{5}{7}$$

- The sequence converges to $\frac{5}{7}$.

2. Pure Math: Determine if the sequence $c_n = \frac{n^2}{3n+1}$ converges or diverges.

- Divide numerator and denominator by the highest power of n in the denominator, which is n :

$$c_n = \frac{\frac{n^2}{n}}{\frac{3n}{n} + \frac{1}{n}} = \frac{n}{3 + \frac{1}{n}}$$

- Take the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} c_n = \frac{\infty}{3 + 0} = \infty$$

- Since the limit is infinity, the sequence **diverges**.

3. Data Science: A machine learning model's loss over n epochs is modeled by the sequence $L_n = 2 + \frac{10}{n^2}$. As training goes on infinitely ($n \rightarrow \infty$), what is the theoretical minimum loss the model converges to?

- Take the limit of L_n as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} \left(2 + \frac{10}{n^2} \right)$$

- By limit laws, this is $\lim_{n \rightarrow \infty} 2 + \lim_{n \rightarrow \infty} \frac{10}{n^2} = 2 + 0 = 2$.
- The theoretical minimum loss is 2.

4. Pure Math: Find the average rate of change of the function $f(x) = x^3$ over the interval $[1, 3]$.

- Use the average rate of change formula: $\frac{f(x_2) - f(x_1)}{x_2 - x_1}$.
- $f(3) = 3^3 = 27$.
- $f(1) = 1^3 = 1$.
- Average rate of change = $\frac{27-1}{3-1} = \frac{26}{2} = 13$.

5. Pure Math: Using the limit definition of the tangent slope $m_{\tan} = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$, find the instantaneous slope of $f(x) = x^2$ at the point $x = 4$.

- Here $c = 4$. $f(4) = 4^2 = 16$.
- $f(4 + h) = (4 + h)^2 = 16 + 8h + h^2$.
- Set up the limit: $\lim_{h \rightarrow 0} \frac{(16 + 8h + h^2) - 16}{h}$.
- Simplify the numerator: $\lim_{h \rightarrow 0} \frac{8h + h^2}{h}$.
- Factor out h : $\lim_{h \rightarrow 0} \frac{h(8 + h)}{h} = \lim_{h \rightarrow 0} (8 + h)$.
- Apply the limit as $h \rightarrow 0$: $8 + 0 = 8$.
- The instantaneous slope is 8.

Week 8: Differentiation

Detailed Solutions

1. Pure Math: Differentiate $f(x) = x^3 e^x$ using the Product Rule.

- Let $u(x) = x^3$ and $v(x) = e^x$.
- $u'(x) = 3x^2$ and $v'(x) = e^x$.
- Apply Product Rule: $f'(x) = u'v + uv' = (3x^2)(e^x) + (x^3)(e^x)$.
- Factor out $x^2 e^x$: $f'(x) = x^2 e^x (3 + x)$.

2. Data Science: A machine learning Mean Squared Error cost function for a single parameter w is given by $C(w) = (w - 3)^2 + 4$. Find the critical point w where the gradient (derivative) is zero. Is this a maximum or minimum?

- Find derivative using the Chain Rule: $C'(w) = 2(w - 3)(1) + 0 = 2w - 6$.
- Set derivative to zero to find critical points: $2w - 6 = 0 \implies 2w = 6 \implies w = 3$.
- Check the second derivative: $C''(w) = 2$.
- Since $C''(3) = 2 > 0$, the function is concave up, meaning $w = 3$ is a **minimum**.

3. Pure Math: Find the derivative of $g(x) = \ln(3x^4 - 2x)$ using the Chain Rule.

- Let the outer function be $\ln(u)$ and the inner function be $u = 3x^4 - 2x$.
- The derivative of the outer function is $\frac{1}{u}$. The derivative of the inner function is $12x^3 - 2$.
- Apply the Chain Rule: $g'(x) = \frac{1}{3x^4 - 2x} \cdot (12x^3 - 2) = \frac{12x^3 - 2}{3x^4 - 2x}$.

4. Pure Math: Use L'Hôpital's Rule to evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.

- Direct substitution gives $\frac{\sin(0)}{0} = \frac{0}{0}$, an indeterminate form.
- Differentiate the numerator: $\frac{d}{dx}(\sin(x)) = \cos(x)$.
- Differentiate the denominator: $\frac{d}{dx}(x) = 1$.
- Take the limit of the new fraction: $\lim_{x \rightarrow 0} \frac{\cos(x)}{1} = \cos(0) = 1$.
- The limit is 1.

5. Data Science: A simple neural network maps input x through two layers: $y = (2x + 1)^3$. Using the Chain Rule, what is $\frac{dy}{dx}$?

- Let the outer function be u^3 and the inner layer be $u = 2x + 1$.
- Derivative of the outer function is $3u^2$. Derivative of the inner function is 2.
- Apply the Chain Rule: $\frac{dy}{dx} = 3(2x + 1)^2 \cdot 2 = 6(2x + 1)^2$.
- This is exactly how backpropagation calculates gradients to update weights!

Week 9: Integration

Detailed Solutions

1. Pure Math: Evaluate the indefinite integral: $\int (4x^3 - \frac{1}{x} + e^x) dx$.

- Apply the Sum Rule to integrate term by term.
- $\int 4x^3 dx = 4 \left(\frac{x^4}{4} \right) = x^4$.
- $\int -\frac{1}{x} dx = -\ln|x|$.
- $\int e^x dx = e^x$.
- Combine and add the integration constant: $x^4 - \ln|x| + e^x + C$.

2. Pure Math: Use u-substitution to evaluate the integral: $\int x^2(x^3 + 5)^4 dx$.

- Let $u = x^3 + 5$. Then $du = 3x^2 dx$.
- We have $x^2 dx$ in the integral, so we can write $\frac{1}{3} du = x^2 dx$.
- Substitute into the integral: $\int u^4 \left(\frac{1}{3} \right) du = \frac{1}{3} \int u^4 du$.
- Integrate with respect to u : $\frac{1}{3} \left(\frac{u^5}{5} \right) + C = \frac{u^5}{15} + C$.
- Substitute back $u = x^3 + 5$: $\frac{(x^3+5)^5}{15} + C$.

3. Data Science: The Probability Density Function of a uniform distribution is given by $p(x) = 0.5$ on the interval $[0, 2]$. Verify that the total probability over this entire interval is exactly 1 by calculating $\int_0^2 0.5 dx$.

- Find the antiderivative of 0.5: $F(x) = 0.5x$.
- Evaluate from 0 to 2: $F(2) - F(0) = (0.5 \cdot 2) - (0.5 \cdot 0)$.
- $1 - 0 = 1$. The total probability is exactly 1.

4. Pure Math: Evaluate the definite integral using FTC Part 2: $\int_1^2 \frac{1}{x^2} dx$.

- Rewrite the integrand as x^{-2} .
- Find the antiderivative using the Power Rule: $\int x^{-2} dx = \frac{x^{-1}}{-1} = -\frac{1}{x}$.
- Evaluate from 1 to 2: $\left[-\frac{1}{2}\right] - \left[-\frac{1}{1}\right]$.
- $-\frac{1}{2} + 1 = \frac{1}{2}$.

5. Pure Math: Find the area of the region enclosed between the curves $y = x^2$ and $y = x$ on the interval $[0, 1]$.

- On the interval $(0, 1)$, $x > x^2$, so the top curve is $f(x) = x$ and the bottom is $g(x) = x^2$.
- Set up the integral: $\int_0^1 (x - x^2) dx$.
- Antiderivative: $\frac{x^2}{2} - \frac{x^3}{3}$.
- Evaluate at limits: $(\frac{1}{2} - \frac{1}{3}) - (0 - 0)$.
- Common denominator is 6: $\frac{3}{6} - \frac{2}{6} = \frac{1}{6}$.
- The area is $\frac{1}{6}$.

Week 10: Graph Theory (I)

Detailed Solutions

1. Computer Science: An undirected graph has 6 vertices and 15 edges. Is it a simple graph, or must it contain multiple edges/loops?

- The maximum number of edges in a simple undirected graph is given by the combination formula $\binom{n}{2} = \frac{n(n-1)}{2}$.
- For $n = 6$, $\max \text{ edges} = \frac{6 \times 5}{2} = 15$.
- Because the graph has exactly 15 edges, it is fully connected (a complete graph K_6). It can be a simple graph (it does not require multiple edges or loops to fit 15 edges).

2. Data Structure: A graph is represented by an Adjacency Matrix. What does it mean if the sum of all elements in the i -th row is 3? Assume the graph is unweighted and directed.

- In a directed adjacency matrix, the i -th row represents all edges **leaving** vertex i .
- Since the graph is unweighted, a 1 represents an edge.
- A sum of 3 means there are exactly 3 outgoing edges from vertex i . This is known as the **out-degree** of vertex i .

3. Algorithms: You are traversing a graph to find the shortest path out of a maze. The graph is unweighted. Should you use BFS or DFS, and why?

- You should use **BFS (Breadth-First Search)**.
- BFS guarantees finding the shortest path in an unweighted graph because it explores layer by layer. DFS will wander deeply into a path and might find a very long, convoluted way out before ever checking a much shorter branch.

4. Algorithms: You have a DAG representing university course prerequisites. You want to generate a valid semester-by-semester graduation plan. Which graph algorithm should you use?

- You should use **Topological Sorting**.
- This algorithm takes a Directed Acyclic Graph and returns a linear sequence of nodes where every prerequisite comes before its dependent course, guaranteeing a valid graduation plan.

5. Complexity: An undirected graph $G = (V, E)$ has $|V| = 10,000$ and $|E| = 20,000$. Which storage representation is more memory efficient: an Adjacency Matrix or an Adjacency List? Justify mathematically.

- **Adjacency Matrix:** Requires $|V|^2$ entries. $10,000^2 = 100,000,000$ entries.
- **Adjacency List:** Requires roughly $O(|V| + 2|E|)$ space for an undirected graph. $10,000 + 40,000 = 50,000$ entries.
- The graph is extremely sparse ($50,000 \ll 100,000,000$). The **Adjacency List** is far more efficient, saving 99.95% of the memory.

Week 11: Graph Theory (II)

Detailed Solutions

1. Algorithms: You are writing navigation software for a delivery drone. The drone battery drains faster if it flies into a headwind, which is modeled as an edge with a negative weight (cost). Which shortest-path algorithm *must* you use?

- You must use the **Bellman-Ford Algorithm**.
- Dijkstra's Algorithm will fail because its greedy logic assumes that adding more edges to a path can only increase the total cost. It cannot handle negative edge weights.

2. Algorithms: An undirected graph has 8 vertices. How many edges will its Minimum Spanning Tree have?

- By definition, any spanning tree of a connected graph with $|V|$ vertices will have exactly $|V| - 1$ edges.
- Therefore, the MST will have $8 - 1 = 7$ edges.

3. Pure Math: You apply Kruskal's algorithm to a graph. The next lightest edge connects Node A and Node B. However, Node A and Node B are already in the same connected component of your growing MST. Should you add the edge? Why or why not?

- You should **not** add the edge.
- If two nodes are already in the same connected component, it means there is already a path between them. Adding a direct edge between them would create a **cycle**, which violates the definition of a tree.

4. Complexity: Why is Floyd-Warshall ($O(|V|^3)$) preferred over running Dijkstra's algorithm from every single node ($O(|V| \cdot |E| \log |V|)$) for dense graphs when finding all-pairs shortest paths?

- In a dense graph, the number of edges $|E|$ approaches $|V|^2$.
- Substituting $|E| = |V|^2$ into the repeated Dijkstra complexity gives $O(|V| \cdot |V|^2 \log |V|) = O(|V|^3 \log |V|)$.
- $O(|V|^3)$ (Floyd-Warshall) is faster than $O(|V|^3 \log |V|)$ (Repeated Dijkstra). Additionally, Floyd-Warshall is much simpler to implement and handles negative weights, whereas Dijkstra does not.

5. Graph Theory: What mathematical operation on an Adjacency Matrix A calculates the number of walks of exactly length k between any two nodes?

- You take the matrix A to the power of k (i.e., compute A^k).
- The value in row i , column j of the resulting matrix A^k is exactly the number of walks of length k from node i to node j .

Week 12: Comprehensive Final Review

Detailed Solutions

1. Sets & Logic: In a dataset of 100 users, 60 like Apples (A), 50 like Bananas (B), and 20 like neither. How many users like *both* Apples and Bananas?

- Total users $N = 100$. Users who like neither is 20.
- Therefore, the number of users who like at least one fruit is $|A \cup B| = 100 - 20 = 80$.
- Using the Principle of Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$.
- $80 = 60 + 50 - |A \cap B| \implies 80 = 110 - |A \cap B|$.
- $|A \cap B| = 110 - 80 = 30$. Exactly 30 users like both.

2. Functions & Exponentials: Given $f(x) = e^{2x}$ and $g(x) = \ln(x)$, find the composite function $(f \circ g)(x)$ and simplify it completely.

- Substitute $g(x)$ into $f(x)$: $(f \circ g)(x) = f(\ln(x)) = e^{2\ln(x)}$.
- Apply the logarithm power rule: $2\ln(x) = \ln(x^2)$.
- The expression becomes $e^{\ln(x^2)}$.
- Using the inverse property of exponentials and logarithms ($e^{\ln(y)} = y$), the simplified function is exactly x^2 (for $x > 0$).

3. Calculus: Find the critical points of the loss function $L(w) = w^3 - 3w^2 - 9w + 15$. Use the Second Derivative Test to determine which point is a local minimum (the optimal weight).

- Find the first derivative: $L'(w) = 3w^2 - 6w - 9$.
- Set to zero: $3(w^2 - 2w - 3) = 0 \implies 3(w - 3)(w + 1) = 0$.
- The critical points are $w = 3$ and $w = -1$.
- Find the second derivative: $L''(w) = 6w - 6$.
- Test $w = -1$: $L''(-1) = 6(-1) - 6 = -12$. Since this is negative, $w = -1$ is a **maximum**.
- Test $w = 3$: $L''(3) = 6(3) - 6 = 12$. Since this is positive, $w = 3$ is the **local minimum**.

4. Integration: Calculate the total probability area under the curve $p(x) = 3x^2$ between $x = 0$ and $x = 1$.

- Set up the definite integral: $\int_0^1 3x^2 dx$.
- Find the general antiderivative using the power rule: $\frac{3x^3}{3} = x^3$.
- Evaluate at the limits (FTC Part 2): $F(1) - F(0) = 1^3 - 0^3 = 1 - 0 = 1$.
- The total probability area is exactly 1.

5. Graph Theory: You are using Dijkstra's algorithm to find the fastest driving route. You discover a toll road that pays you \$5 to drive on it (a negative weight of -5). Will Dijkstra's algorithm still guarantee the correct shortest path? What algorithm should you use instead?

- **No**, Dijkstra's algorithm fundamentally assumes all edge weights are non-negative. It operates on a greedy strategy, locking in the shortest distance to a node and never looking back. A negative weight would invalidate previously locked distances.
- You should use the **Bellman-Ford algorithm**, which systematically relaxes all edges multiple times and correctly handles negative edge weights.